

HSC/Alim 2026

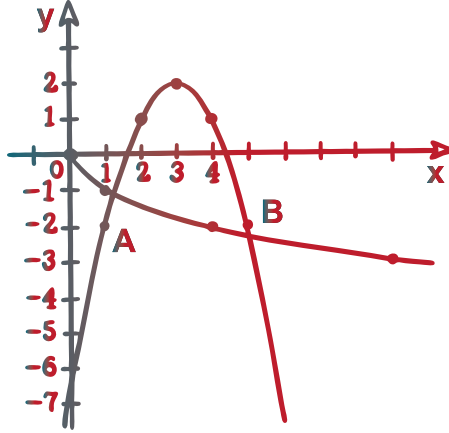
২০২৬ পরীক্ষার্থীদের জন্য

ফাইনাল রিভিশন কোর্স

প্র্যাকটিস শীট

উচ্চতর গণিত ১ম পত্র

অধ্যায়-১০ : যোগজীকরণ



উদ্ভাস

একাডেমিক এন্ড এডমিশন কেয়ার



অধ্যায় ১০

যোগজীকরণ

লেকচার-২১

টাইপভিত্তিক বিগত ৭ বছরের (২০১৯-২০২৫) MCQ প্রশ্নের অ্যানালাইসিস:

গুরুত্ব	টাইপের নাম	DB	RB	Ctg.B	BB	SB	JB	CB	Din.B	MB	Mad.B	মোট
☆☆	T-01: যোগজীকরণের সাধারণ সূত্র $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ব্যবহার করে	1	1	2	-	1		1	1		2	9
☆	T-02: $\int f(ax+b) dx$ আকৃতির	2	-	-	-	-	-	-	-	-	-	2
☆☆☆	T-03: প্রতিস্থাপন পদ্ধতিতে 'z' ধরে যোগজীকরণ	12	10	13	7	5	8	8	10	5	3	81
☆☆	T-04: $\int \frac{f'(x)}{f(x)} dx$ এবং $\int \frac{f'(x)}{\sqrt{f(x)}} dx$ আকৃতির	1	-	-	1		2	2	2	-	-	8
	T-05: Exponential আকৃতির	-	-	-	-	-	-	-	-	-	-	-
☆	T-06: $\int \sin ax \cdot \cos bx dx$, $\int \sin ax \cdot \sin bx dx$, $\int \cos ax \cdot \cos bx dx$ আকৃতির	-	1	-	-	-	-	-	-	1	-	2
☆☆	T-07: $\int \frac{dx}{1+\sin ax}$, $\int \frac{dx}{1-\sin ax}$, $\int \frac{dx}{1+\cos ax}$, $\int \frac{dx}{1-\cos ax}$ আকারের	1	2	1	2	2	1	1	1	-	-	11
☆	T-08: $\int \sin^m x dx$; $\int \cos^m x dx$ $\int \sin^m x \cdot \cos^n x dx$ আকারের	-	1	2	-	-	1	1	-	1	-	6

MCQ প্রশ্ন

- $\int \frac{3}{2-5x} dx = f(x) + c$ হলে $f(x)$ এর মান কোনটি? [DB'25]
 - $\frac{3}{5} \ln(2-5x)$
 - $-\frac{3}{5} \ln(2-5x)$
 - $3 \ln(2-5x)$
 - $-3 \ln(2-5x)$
- $\int \frac{1}{1+\sin x} dx =$ কত? [BB'25]
 - $(\tan x - \sec x) + c$
 - $(\sec x - \tan x) + c$
 - $(\tan x + \sec x) + c$
 - $-(\tan x + \sec x) + c$
- $\int \cos bx dx =$ কত? [BB'25]
 - $-\frac{1}{b} \sin bx + c$
 - $\frac{1}{b} \sin bx + c$
 - $-b \sin bx + c$
 - $b \sin bx + c$
- $\int \tan x dx$ এর মান- [SB'25; MB'23]
 - $-\ln|\cos x| + c$
 - $\ln|\sec x| + c$
 - $\ln|\cot x| + c$
 নিচের কোনটি সঠিক?
 - i, ii
 - i, iii
 - ii, iii
 - i, ii, iii
- দি $\int P^x dx = F(x) + c$ হয়, তবে $F(x)$ এর মান কোনটি? [DB'23]
 - $\frac{P^x}{\log_e P}$
 - $P^x \log_e P$
 - xP^{x-1}
 - $\frac{P^{x+1}}{x+1}$
- দি $\int \operatorname{cosec}^2(5-8x) dx = a \cot(5-8x) + c$ হয়, তবে a এর মান কোনটি? [DB'23]
 - 8
 - $-\frac{1}{8}$
 - $\frac{1}{8}$
 - 8
- $\int \cos^2 3x dx =$ কত? [Ctg.B'23]
 - $\frac{1}{2}(x + \sin 6x) + c$
 - $\frac{1}{2}(x - \sin 6x) + c$
 - $\frac{1}{12}(6x + \sin 6x) + c$
 - $\frac{1}{12}(6x - \sin 6x) + c$
- $\int \frac{1}{x^2} dx =$ কত? [$x \neq 0$] [Ctg.B'23]
 - $-\frac{1}{3x^3} + c$
 - $-\frac{1}{x} + c$
 - $\frac{1}{3x^3} + c$
 - $\frac{1}{x} + c$

MCQ উত্তরমালা

01. b	02. a	03. b	04. a	05. a	06. c	07. c	08. b
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09. $\int \frac{1+\tan^2 x}{(1-\tan x)^2} dx$ এর মান কোনটি? [Din.B'23]
 (a) $\frac{-1}{1+\tan x} + c$ (b) $\frac{-1}{1-\tan x} + c$
 (c) $\frac{1}{1-\tan x} + c$ (d) $\frac{1}{1+\tan x} + c$
10. y -অক্ষ এবং $x = 4 - y^2$ পরাবৃত্ত দ্বারা আবদ্ধ ক্ষেত্রের ক্ষেত্রফল কত? [SB'22]
 (a) $\frac{3}{32}$ বর্গ একক (b) $\frac{2}{33}$ বর্গ একক
 (c) $\frac{32}{3}$ বর্গ একক (d) $\frac{33}{2}$ বর্গ একক
11. $\int 10^{5x} dx =$ কত? [BB'22; Ctg'21]
 (a) $\frac{10^{5x}}{\ln 10} + c$ (b) $\frac{10^{5x}}{5 \ln 10} + c$
 (c) $5 \cdot 10^{5x} \ln 10 + c$ (d) $10^{5x} \ln 10 + c$
12. $\int \frac{5}{1-5x} dx$ এর মান কোনটি? [Din.B'22; SB'21]
 (a) $\ln(1-5x) + c$ (b) $-\ln(1-5x) + c$
 (c) $\frac{\ln(1-5x)}{5}$ (d) $\frac{-\ln(1-5x)}{5}$
13. $\int \sin\left(10 - \frac{x}{5}\right) dx =$ কত? [DB'21]
 (a) $5 \cos\left(10 - \frac{x}{5}\right) + c$ (b) $-5 \cos\left(10 - \frac{x}{5}\right) + c$
 (c) $-\frac{1}{5} \cos\left(10 - \frac{x}{5}\right) + c$ (d) $\frac{1}{5} \cos\left(10 - \frac{x}{5}\right) + c$
- নিচের উদ্দীপকের আলোকে পরবর্তী প্রশ্নের উত্তর দাও:
 $f(x) = \cot x, g(x) = \operatorname{cosec}^2 x$
14. $\int f(x) dx =$ কত? [DB'21]
 (a) $\operatorname{cosec}^2 x + c$ (b) $-\operatorname{cosec}^2 x + c$
 (c) $\ln|\operatorname{cosec} x| + c$ (d) $\ln(\sin x) + c$
15. নিচের তথ্যগুলো লক্ষ কর: [DB'21]
 (i) $\int g\left(\frac{x}{2}\right) dx = -2 \cot \frac{x}{2} + c$
 (ii) $\frac{d}{dx} \left\{ f\left(\frac{\pi}{2} - 2x\right) \right\} = 2 \sec^2 2x$
 (iii) $\int f(x) g(x) dx = -\frac{1}{2} \cot^2 x + c$
 নিচের কোনটি সঠিক?
 (a) i, ii (b) i, iii (c) ii, iii (d) i, ii, iii
16. $\int \sin x^\circ dx =$ কত? [Ctg.B'21; DB'17]
 (a) $-\cos x^\circ + c$ (b) $\cos x^\circ + c$
 (c) $-\frac{180}{\pi} \cos \frac{\pi x}{180} + c$ (d) $\frac{180}{\pi} \cos \frac{\pi x}{180} + c$
17. $\int \frac{dx}{1-\cos x} = f(x) + c$ হলে $f(x)$ এর মান কোনটি? [SB'21; RB'19; Ail.B'18]
 (a) $-\cot \frac{x}{2}$ (b) $-2 \cot \frac{x}{2}$ (c) $2 \cot \frac{x}{2}$ (d) $\cot \frac{x}{2}$
18. $\int \sqrt{2-3x} dx =$ কত? [BB'21]
 (a) $-\frac{2}{9} (2-3x)^{\frac{3}{2}} + c$ (b) $-\frac{1}{6} (2-3x)^{-\frac{1}{2}} + c$
 (c) $-(2-3x)^{\frac{3}{2}} + c$ (d) $-3(2-3x)^{-\frac{1}{2}} + c$
19. $\int \frac{4x^3}{1+x^8} dx = f(x) + c$ হলে $f(x)$ এর মান কত? [Din.B'21]
 (a) $\frac{1}{1+x^2}$ (b) $\sin^{-1} x^3$ (c) $\tan^{-1} x^4$ (d) $\tan^{-1} x^3$
20. $\int x e^{x^2} dx = ? + c$ [Ctg.B'19]
 (a) $2e^{x^2}$ (b) e^{x^2} (c) $x e^{x^2}$ (d) $\frac{1}{2} e^{x^2}$
21. $\int \frac{\sin 2x}{\cos^2 2x} dx =$ কত? [BB'19]
 (a) $\frac{1}{2} \sec 2x + c$ (b) $2 \sec 2x + c$
 (c) $\frac{1}{2} \operatorname{cosec} 2x + c$ (d) $2 \operatorname{cosec} 2x + c$
22. $f(x) = \sin 2x, g(x) = \sin^2 x$ এর প্রেক্ষিতে $\frac{f(x)}{g(x)}$ এর অনির্দিষ্ট যোগজ কোনটি? [CB'17]
 (a) $2 \ln|1 + \cos 2x| + c$ (b) $-\ln|1 - \cos 2x| + c$
 (c) $\ln|1 + \cos 2x| + c$ (d) $\ln|1 - \cos 2x| + c$
23. $\int \cos^2 2x dx =$ কোনটি?
 (a) $\frac{1}{2} \left(x + \frac{1}{4} \sin 4x\right) + c$ (b) $\frac{1}{3} \left(x + \frac{1}{6} \sin 4x\right) + c$
 (c) $\frac{1}{2} \left(x - \frac{1}{4} \sin 2x\right) + c$ (d) $\frac{1}{2} \left(x + \frac{1}{4} \sin 2x\right) + c$
24. $\int \frac{(\ln x)^2}{x} dx$ এর মান নিচের কোনটি?
 (a) $\ln x^3$ (b) $\ln x^2$
 (c) $\ln x$ (d) $3^{-1} (\ln x)^3$

MCQ উত্তরমালা

09. c	10. c	11. b	12. b	13. a	14. d	15. d	16. c
17. a	18. a	19. c	20. d	21. a	22. d	23. a	24. d

MCQ প্রশ্নের ব্যাখ্যামূলক সমাধান

01. $\int \frac{3}{2-5x} dx = \frac{-3}{5} \int \frac{-5}{2-5x} dx = -\frac{3}{5} \int \frac{d[2-5x]}{2-5x} dx$
 $= -\frac{3}{5} \ln|2-5x| + c \therefore f(x) = -\frac{3}{5} \ln|2-5x|$
02. $\int \frac{1}{1+\sin x} dx = \int \frac{1-\sin x}{1-\sin^2 x} dx = \int \frac{1-\sin x}{\cos^2 x} dx$
 $= \int (\sec^2 x - \sec x \tan x) dx$
 $= \tan x - \sec x + c$
03. $\int \cos bx dx = \frac{\sin bx}{b} + c = \frac{1}{b} \sin bx + c$

04.

$$\begin{aligned} \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{-dz}{z} \\ &= -\int \frac{1}{z} dz = -\ln|z| + c \\ &= -\ln|\cos x| + c \quad [\because (i) \text{ নং সঠিক}] \\ &= \ln\left|\frac{1}{\cos x}\right| + c = \ln|\sec x| + c \\ &[\because (ii) \text{ নং সঠিক}] \\ \therefore \int \tan x dx &\neq \ln|\cot x| + c \\ \therefore (iii) \text{ নং সঠিক নয়} \end{aligned}$$

ধরি,
 $z = \cos x$
 $dz = -\sin x dx$





$$05. \frac{d}{dx}(P^x) = \frac{P^x \log_e P}{1} = \int P^x \cdot \log_e P \, dx = P^x + c$$

$$\Rightarrow \log_e P \times \int P^x \, dx = P^x + c$$

$$\Rightarrow \int P^x \, dx = \frac{P^x}{\log_e P} + c = F(x) + c$$

$$\therefore F(x) = \frac{P^x}{\log_e P}$$

$$06. \int \operatorname{cosec}^2(5-8x) \, dx = -\frac{1}{8} \int \operatorname{cosec}^2(5-8x) \, d(5-8x)$$

$$= -\frac{1}{8} \cot(5-8x) + c$$

$$\text{এখন, } \frac{1}{8} \cot(5-8x) + c = a \cot(5-8x) + c \text{ হতে পাই, } a = \frac{1}{8}$$

$$07. \int \cos^2 3x \, dx = \frac{1}{2} \int 2\cos^2 3x \, dx$$

$$= \frac{1}{2} \int (1 + \cos 6x) \, dx = \frac{1}{2} \left[x + \frac{\sin 6x}{6} \right] + c$$

$$= \frac{1}{2} \left(\frac{6x + \sin 6x}{6} \right) + c = \frac{1}{12} (6x + \sin 6x) + c$$

$$08. \int \frac{1}{x^2} \, dx = \int x^{-2} \, dx = \frac{x^{-2+1}}{-2+1} + c = \frac{x^{-1}}{-1} + c = -\frac{1}{x} + c$$

$$09. \text{ধরি, } u = 1 - \tan x$$

$$du = -\sec^2 x \, dx; -du = \sec^2 x \, dx$$

$$\int \frac{1+\tan^2 x}{(1-\tan x)^2} \, dx = \int \frac{\sec^2 x}{(1-\tan x)^2} \, dx = -\int \frac{du}{u^2}$$

$$= -\frac{u^{-2+1}}{-2+1} + c = \frac{1}{u} + c = \frac{1}{1-\tan x} + c$$

$$10. y^2 = 4-x \Rightarrow y^2 = -(x-4)$$

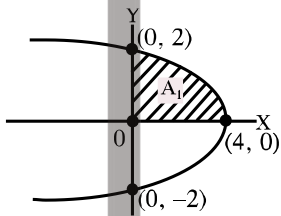
$$y \text{ অক্ষের ছেদবিন্দু, } x = 0$$

$$\therefore 4 - y^2 = 0 \Rightarrow y = \pm 2$$

$$\therefore \text{নির্ণেয় ক্ষেত্রফল} = 2 \int_0^2 x \, dy = 2 \int_0^2 (4 - y^2) \, dy$$

$$= 2 \left[4y - \frac{y^3}{3} \right]_0^2 = 2 \left[8 - \frac{8}{3} - 0 \right] \text{ বর্গ একক}$$

$$= \frac{32}{3} \text{ বর্গ একক।}$$



$$11. \int 10^{5x} \, dx$$

$$\text{ধরি, } 5x = z \Rightarrow 5 = \frac{dz}{dx} \Rightarrow dx = \frac{dz}{5}$$

$$\therefore \frac{1}{5} \int 10^z \, dz = \frac{1}{5} \cdot \frac{10^z}{\ln 10} + c = \frac{10^{5x}}{5 \ln 10} + c$$

$$12. \int \frac{5}{1-5x} \, dx = \frac{5 \ln|1-5x|}{-5} + c = -\ln|1-5x| + c$$

$$13. 10 - \frac{x}{5} = y \Rightarrow -\frac{dx}{5} = dy \Rightarrow dx = -5dy$$

$$\therefore \int \sin \left(10 - \frac{x}{5} \right) dx = \int \sin y (-5 \, dy)$$

$$= 5 \cos y + c = 5 \cos \left(10 - \frac{x}{5} \right) + c$$

$$14. \int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx = \ln|\sin x| + c$$

$$\left[\because \int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + c \right]$$

$$15. (i) \int \operatorname{cosec}^2 \left(\frac{x}{2} \right) dx = -\frac{\cot \left(\frac{x}{2} \right)}{\frac{1}{2}} + c$$

$$= -2 \cot \left(\frac{x}{2} \right) + c$$

$$(ii) \frac{d}{dx} \left\{ \cot \left(\frac{\pi}{2} - 2x \right) \right\} = \frac{d}{dx} (\tan 2x) = 2 \sec^2 2x$$

$$(iii) \int \operatorname{cosec}^2 x \cot x \, dx$$

$$= -\int \cot x (-\operatorname{cosec}^2 x) \, dx$$

$$= -\int z \, dz = -\frac{z^2}{2} + c$$

$$= -\frac{1}{2} \cot^2 x + c$$

$$16. x^\circ = \left(\frac{\pi x}{180} \right)^\circ \therefore \int \sin x^\circ \, dx$$

$$= \int \sin \frac{\pi x}{180} \, dx = \frac{-\cos \frac{\pi x}{180}}{\frac{\pi}{180}} = -\frac{180}{\pi} \cos \frac{\pi x}{180} + c$$

$$17. \int \frac{dx}{2 \sin^2 \frac{x}{2}} = \frac{1}{2} \int \operatorname{cosec}^2 \frac{x}{2} \, dx$$

$$= \frac{1}{2} \left[-\frac{\cot \frac{x}{2}}{\frac{1}{2}} \right] + c = -\cot \frac{x}{2} + c$$

$$18. \int (2-3x)^{\frac{1}{2}} dx = -\frac{1}{3} \times \frac{2}{3} (2-3x)^{\frac{3}{2}} + c$$

$$= -\frac{2}{9} (2-3x)^{\frac{3}{2}} + c$$

$$19. \int \frac{4x^3}{1+(x^4)^2} \, dx = \int \frac{dz}{1+z^2} = \tan^{-1}(z) + c$$

$$= \tan^{-1}(x^4) + c \quad [\text{ধরি, } x^4 = z \Rightarrow 4x^3 \, dx = dz]$$

$$20. \text{ধরি, } x^2 = z$$

$$\Rightarrow 2x \, dx = dz \Rightarrow x \, dx = \frac{dz}{2}$$

$$\frac{1}{2} \int e^z \, dz = \frac{1}{2} e^z + c = \frac{1}{2} e^{x^2} + c$$

$$21. \int \frac{\sin 2x}{\cos^2 2x} \, dx = \int \tan 2x \cdot \sec 2x \, dx = \frac{1}{2} \sec 2x + c$$

$$22. \int \frac{\sin 2x}{\sin^2 x} \, dx = \int \frac{2 \sin x \cdot \cos x}{\sin^2 x} \, dx \left[\int \frac{f'(x)}{f(x)} \, dx \right]$$

$$= \ln|\sin^2 x| + c = \ln \left| \frac{1-\cos 2x}{2} \right| + c$$

$$= \ln|1-\cos 2x| - \ln 2 + c = \ln|1-\cos 2x| + c$$

$$23. \int \cos^2 2x \, dx = \frac{1}{2} \int 2 \cos^2 2x \, dx = \frac{1}{2}$$

$$\int (1 + \cos 4x) \, dx = \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$$

$$24. \int \frac{(\ln x)^2}{x} \, dx = \int u^2 \, du$$

$$= \frac{u^3}{3} = \frac{(\ln x)^3}{3} = 3^{-1} (\ln x)^3 \quad \left| \text{ধরি, } \ln x = u \Rightarrow \frac{1}{x} \, dx = du \right|$$





❖ টাইপভিত্তিক বিগত ৭ বছরের (২০১৯-২০২৫) সৃজনশীল (ক, খ, গ) প্রশ্নের অ্যানালাইসিস:

গুরুত্ব	টাইপের নাম	যতবার প্রশ্ন এসেছে			DB	RB	Ctg.B	BB	SB	JB	CB	Din.B	MB	Mad.B	মোট
		(ক)	(খ)	(গ)											
☆	T-01: যোগজীকরণের সাধারণ সূত্র $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ব্যবহার করে	2		-	-	1	-	-	-	-	-	-	-	-	2
☆	T-02: $\int f(ax + b) dx$ আকৃতির	3	1	-	-	2	1	-	-	-	-	-	1	-	4
☆☆	T-03: প্রতিস্থাপন পদ্ধতিতে 'z' ধরে যোগজীকরণ	6	2	1		2		1	1	1	3	1	-	1	10
☆	T-04: $\int \frac{f'(x)}{f(x)} dx$ এবং $\int \frac{f'(x)}{\sqrt{f(x)}} dx$ আকৃতির	4	-	-	-	-	-	2	1	-	-	1	-	-	4
☆	T-05: Exponential আকৃতির	4	-	-	-	1	1	1	-	-	1	-	-	-	4
☆	T-06: $\int \sin ax \cdot \cos bx dx$, $\int \sin ax \cdot \sin bx dx$, $\int \cos ax \cdot \cos bx dx$ আকৃতির	2	-	-	-	-	1	-	-	1	-	-	-	-	2
☆	T-07: $\int \frac{dx}{1+\sin ax}$, $\int \frac{dx}{1-\sin ax}$, $\int \frac{dx}{1+\cos ax}$, $\int \frac{dx}{1-\cos ax}$ আকারের	1	1	-	-	-	-	-	-	1	1	-	-	-	2
☆	T-08: $\int \sin^m x dx$; $\int \cos^m x dx$; $\int \sin^m x \cdot \cos^n x dx$ আকারের	1	1	-	-	-	1	-	1	-	-	-	-	-	2

CQ প্রশ্ন

01. দৃশ্যকল্প-১: $f(x) = \sin x$ [DB'25]
 (ক) $\int_1^3 \frac{dx}{x(2+\ln x)}$ নির্ণয় কর।
 (খ) $\int_0^2 \frac{f'(x)dx}{\{2+f(x)\}\{3+f(x)\}}$ এর মান দৃশ্যকল্প-১ ব্যবহার করে নির্ণয় কর।
02. (ক) দেখাও যে, $\int \tan x dx = \ln|\sec x| + c$ [BB'25]
03. (ক) $\int \frac{1}{1+\tan x} dx$ নির্ণয় কর। [Din.B'25]
04. (ক) $\int \frac{dx}{e^x + e^{-x}}$ এর যোগজ নির্ণয় কর। [CB'23]
05. $h(x) = \tan x$ [CB'22]
 (ক) $\int \frac{h'(x)}{\{1+h(x)\}^2} dx$ নির্ণয় কর।
06. (ক) $\int \frac{5e^{2x}}{1+e^{4x}} dx$ নির্ণয় কর। [RB'21]
07. $f(x) = \sin x - \cos x$ [RB'21]
 (ক) $\int \{f(x)\}^2 dx$ নির্ণয় কর।
 (খ) $\int \frac{1-f(x)}{1+f(x)} dx$ নির্ণয় কর।
08. (ক) যোগজ নির্ণয় কর: $\int xe^{x^2} dx$. [CB'21]
09. (ক) $\int \sin^{11} x \cos x dx$ নির্ণয় কর। [SB'21]
10. (ক) $\int \sin 5x \sin 3x dx$ নির্ণয় কর।
11. (ক) $\int \frac{\sin^3 x - \cos^3 x}{\sin^2 x \cos^2 x} dx$ এর মান নির্ণয় কর।

CQ প্রশ্নের সমাধান

01. (ক) Solⁿ: $\int_1^3 \frac{dx}{x(2+\ln x)} = \int_2^5 \frac{dz}{z}$
 $= [\ln z]_2^5 = \ln 5 - \ln 2$ ধরি, $2 + \ln x = z \Rightarrow \frac{1}{x} dx = dz$
 $= \ln \left| \frac{5}{2} \right|$ (Ans.)
- | | | |
|---|---|-------|
| x | 1 | e^3 |
| z | 2 | 5 |
- (খ) Solⁿ: দৃশ্যকল্প-১ হতে পাই, $f(x) = \sin x$
 $\therefore f'(x) = \frac{d}{dx}(\sin x) = \cos x$
 $\therefore \int_0^2 \frac{f'(x)dx}{\{2+f(x)\}\{3+f(x)\}} = \int_0^2 \frac{\cos x dx}{(2+\sin x)(3+\sin x)}$
 $= \int_0^2 \left(\frac{\cos x}{2+\sin x} - \frac{\cos x}{3+\sin x} \right) dx$
 $= \int_0^2 \frac{\cos x dx}{2+\sin x} - \int_0^2 \frac{\cos x dx}{3+\sin x}$
- এখন, $\int_0^2 \frac{\cos x dx}{2+\sin x} = \int_2^5 \frac{du}{u} = [\ln|u|]_2^5 = \ln 5 - \ln 2 = \ln \frac{5}{2}$
- আবার, $\int_0^2 \frac{\cos x dx}{3+\sin x} = \int_3^4 \frac{dv}{v} = [\ln|v|]_3^4 = \ln 4 - \ln 3 = \ln \frac{4}{3}$
- $\therefore \int_0^2 \frac{f'(x) dx}{\{2+f(x)\}\{3+f(x)\}} = \int_0^2 \frac{\cos x dx}{2+\sin x} - \int_0^2 \frac{\cos x dx}{3+\sin x}$
 $= \ln \frac{5}{2} - \ln \frac{4}{3} = \ln \left(\frac{5}{2} \times \frac{3}{4} \right) = \ln \left(\frac{15}{8} \right)$ (Ans.)
- ধরি, $2 + \sin x = u$
 $\Rightarrow \cos x dx = du$
- | | | |
|---|---|-----------------|
| x | 0 | $\frac{\pi}{2}$ |
| u | 2 | 3 |
- ধরি, $3 + \sin x = v$
 $\Rightarrow \cos x dx = dv$
- | | | |
|---|---|-----------------|
| x | 0 | $\frac{\pi}{2}$ |
| v | 3 | 4 |





02.

(ক) Solⁿ: L. H. S = $\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$
 $= \int \frac{-dz}{z}$ [ধরি, $\cos x = z \Rightarrow -\sin x \, dx = dz$]
 $= -\ln|z| + c = \ln\left|\frac{1}{z}\right| + c = \ln\left|\frac{1}{\cos x}\right| + c$
 $= \ln|\sec x| + c = \text{R. H. S (Showed)}$

03.

(ক) Solⁿ: $\int \frac{1}{1+\tan x} \, dx = \int \frac{1}{1+\frac{\sin x}{\cos x}} \, dx$
 $= \int \frac{\cos x}{\cos x + \sin x} \, dx$
 $= \frac{1}{2} \int \frac{2 \cos x}{\sin x + \cos x} \, dx$
 $= \frac{1}{2} \int \frac{2 \cos x + \sin x - \sin x}{\sin x + \cos x} \, dx$
 $= \frac{1}{2} \int \frac{(\sin x + \cos x) + (\cos x - \sin x)}{(\sin x + \cos x)} \, dx$
 $= \frac{1}{2} \int \left(1 + \frac{\cos x - \sin x}{\sin x + \cos x}\right) \, dx$
 $= \frac{1}{2} [x + \ln|\sin x + \cos x|] + c \text{ (Ans.)}$

04.

(ক) Solⁿ: $\int \frac{dx}{e^x + e^{-x}} = \frac{dx}{e^{-x}(e^{2x} + 1)}$
 $= \int \frac{e^x dx}{(1 + e^{2x})} = \int \frac{dz}{1 + z^2}$ [ধরি, $z = e^x \therefore dz = e^x dx$]
 $= \tan^{-1} z + c = \tan^{-1}(e^x) + c \text{ (Ans.)}$

05.

(ক) Solⁿ: $h(x) = \tan x \therefore h'(x) = \sec^2 x$
 $\therefore \int \frac{h'(x)}{\{1+h(x)\}^2} \, dx = \int \frac{\sec^2 x}{(1+\tan x)^2} \, dx$
 ধরি, $1 + \tan x = z \Rightarrow \sec^2 x \, dx = dz$
 $= \int \frac{dz}{z^2} = -\frac{1}{z} + c = -\frac{1}{1+\tan x} + c \text{ (Ans.)}$

06.

(ক) Solⁿ: $\int \frac{5e^{2x}}{1+e^{4x}} \, dx = \int \frac{5 \cdot \frac{1}{2} du}{1+u^2}$ [ধরি, $e^{2x} = u$]
 $= \frac{5}{2} \int \frac{du}{1+u^2} = \frac{5}{2} \tan^{-1} u + c$
 $= \frac{5}{2} \tan^{-1}(e^{2x}) + c \text{ (Ans.)}$

07.

(ক) Solⁿ: $\int f(x)^2 \, dx = \int (\sin x - \cos x)^2 \, dx$
 $= \int \{(\sin^2 x + \cos^2 x) - 2\sin x \cos x\} \, dx$
 $= \int (1 - \sin 2x) \, dx$
 $= x + \frac{1}{2} \cos 2x + c \text{ (Ans.)}$

(খ) Solⁿ: $\int \frac{1-f(x)}{1+f(x)} \, dx = \int \frac{1-(\sin x - \cos x)}{1+(\sin x - \cos x)} \, dx$
 $= \int \frac{\{1-(\sin x - \cos x)\}^2 \, dx}{\{1+(\sin x - \cos x)\}\{1-(\sin x - \cos x)\}}$
 $= \int \frac{1-2(\sin x - \cos x) + (\sin^2 x + \cos^2 x - \sin 2x)}{1-(\sin x - \cos x)^2} \, dx$
 $= \int \frac{2 - \sin 2x - 2\sin x + 2\cos x}{(1-1) + \sin 2x} \, dx$
 $= \int \{2 \operatorname{cosec} 2x - 1 - \sec x - \operatorname{cosec} x\} \, dx$
 $= \ln(\tan x) - x - \ln|\sec x + \tan x| - \ln \tan \frac{x}{2} + c \text{ (Ans.)}$

08.

(ক) Solⁿ: ধরি, $x^2 = z \Rightarrow 2x \, dx = dz$
 $\therefore \int x e^{x^2} \, dx = \frac{1}{2} \int e^z \, dz = \frac{e^z}{2} + c = \frac{e^{x^2}}{2} + c \text{ (Ans.)}$

09.

(ক) Solⁿ: ধরি, $\sin x = z$
 $\therefore \cos x \, dx = dz$
 $\therefore \int \sin^{11} x \cos x \, dx = \int z^{11} \, dz$
 $= \frac{1}{12} z^{12} + c = \frac{1}{12} \sin^{12} x + c \text{ (Ans.)}$

10.

(ক) Solⁿ: $\int \sin 5x \sin 3x \, dx$
 $= \frac{1}{2} \int 2 \sin 5x \sin 3x \, dx$
 $= \frac{1}{2} \int (\cos 2x - \cos 8x) \, dx$
 $= \frac{1}{2} \left(\frac{\sin 2x}{2} - \frac{\sin 8x}{8} \right) + c$
 $= \frac{1}{16} (4 \sin 2x - \sin 8x) + c \text{ (Ans.)}$

11.

(ক) Solⁿ: $\int \frac{\sin^3 x - \cos^3 x}{\sin^2 x \cos^2 x} \, dx$
 $= \int \left(\frac{\sin^3 x}{\sin^2 x \cos^2 x} - \frac{\cos^3 x}{\sin^2 x \cos^2 x} \right) \, dx$
 $= \int \left(\frac{\sin x}{\cos^2 x} - \frac{\cos x}{\sin^2 x} \right) \, dx$
 $= \int (\tan x \cdot \sec x - \cot x \cdot \operatorname{cosec} x) \, dx$
 $= \int \sec x \tan x \, dx - \int \operatorname{cosec} x \cot x \, dx$
 $= \sec x + \operatorname{cosec} x + c$





লেকচার-২২

টাইপভিত্তিক বিগত ৭ বছরের (২০১৯-২০২৫) MCQ প্রশ্নের অ্যানালাইসিস:

গুরুত্ব	টাইপের নাম	DB	RB	Ctg.B	BB	SB	JB	CB	Din.B	MB	Mad.B	মোট
☆	T-09: $\int \frac{dx}{ax^2+bx+c}$, $\int \frac{dx}{\sqrt{ax^2+bx+c}}$ $\int \sqrt{ax^2+bx+c} dx$ আকারের যেখানে a, b, c যেকোনো ধ্রুবসংখ্যা	2	1	1	-	-	-	-	-	-	-	4
☆☆	T-10: $\int \frac{px+q}{ax^2+bx+c} dx$, $\int \frac{px+q}{\sqrt{ax^2+bx+c}} dx$, $\int (px+q)\sqrt{ax^2+bx+c} dx$ আকারের; যেখানে a, b, c, p, q যেকোনো ধ্রুবক	1	2	1	2	-	1	1	1	-	-	9
☆	T-11: $\int \frac{dx}{(ax+b)\sqrt{cx+d}}$ $\int (ax+b)\sqrt{cx+d} dx$; $\int \frac{ax+b}{\sqrt{cx+d}}$ আকারের	-	-	-	1	1	-	-	-	-	-	2
☆☆	T-12: অংশক্রমে সমাকলন বা $\int uv dx$ সংক্রান্ত	2	1	-	2	1	2	2	2	1	-	11
☆☆	T-13: অংশক্রমে সমাকলনের ক্ষেত্রে: $\int e^{ax} \{af(x) + f'(x)\} dx$ আকৃতির	1	2	2	1	4	1	-	1	2	1	15
☆	T-14: আংশিক ভগ্নাংশের সাহায্যে সমাকলন	-	-	-	-	1	-	1	-	2	-	4

MCQ প্রশ্ন

01. $\int \frac{dx}{\sqrt{x^2-a^2}} =$ কত? [DB'25]
 (a) $\frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + c$ (b) $\frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + c$
 (c) $\ln |x + \sqrt{x^2 - a^2}| + c$ (d) $\sin^{-1} \frac{x}{a} + c$
 নিচের উদ্দীপকের আলোকে পরবর্তী প্রশ্নের উত্তর দাও:
 $f(x) = \sin x$, $g(x) = \cos x$
02. $\int e^x \{f(x) + g(x)\} dx =$ কত? [RB, Ctg.B, JB'25; SB'22; DB'19]
 (a) $e^x \sin x + c$ (b) $e^x \cos x + c$
 (c) $e^x (\cos x + \sin x) + c$ (d) $e^x (\cos x - \sin x) + c$
03. $\int_2^3 \frac{2x-1}{x(x-1)} dx =$ কত? [Ctg.B'25; DB'19; RB'17]
 (a) $\ln 4$ (b) $\ln 2$ (c) $2 \ln 3$ (d) $\ln 3$
04. $\int e^x \left(\frac{1}{x^2} - \frac{1}{x} \right) dx$ এর মান- [MB'25; SB'23]
 (a) $\frac{1}{x} + c$ (b) $-\frac{1}{x} + c$ (c) $e^x \cdot \frac{1}{x} + c$ (d) $-\frac{e^x}{x} + c$
05. $\int e^x (x+1) dx =$ [SB'23]
 (a) $x + e^x + c$ (b) $e^x + x + c$
 (c) $e^x x^2 + c$ (d) $xe^x + c$
06. $\int x \sec^2 x dx =$ কত? [CB'23]
 (a) $x \tan x + \ln |\sin x| + c$ (b) $x \tan x - \ln |\sin x| + c$
 (c) $x \tan x + \ln |\cos x| + c$ (d) $x \tan x - \ln |\cos x| + c$
07. $\int \frac{xdx}{\sqrt{1-x^2}} = f(x) + c$ হলে, $f(x) =$ কত? [Ctg.B, Din.B'23]
 (a) $\sin^{-1} x$ (b) $\cos^{-1} x$
 (c) $\ln(\sqrt{1-x^2})$ (d) $-\sqrt{1-x^2}$
08. $\int \frac{1}{9+x^2} dx$ এর মান কত? [MB'23]
 (a) $\frac{1}{3} \tan^{-1} \frac{x}{3} + c$ (b) $\frac{-1}{3} \tan \frac{x}{3} + c$
 (c) $\frac{1}{3} \tan^{-1} \frac{3}{x} + c$ (d) $\tan^{-1} \frac{x}{3} + c$
09. $\int \frac{xdx}{\sqrt{1-x^2}}$ এর মান কোনটি? [RB'22]
 (a) $\frac{2}{3} (1-x)^{\frac{3}{2}} - 2(1-x)^{\frac{1}{2}} + c$
 (b) $-\frac{2}{3} (\sqrt{1-x})(x+2) + c$
 (c) $\frac{1}{3} (1-x)^{\frac{3}{2}} - (1-x)^{\frac{1}{2}} + c$
 (d) $\frac{2}{3} (\sqrt{1-x})(x+2) + c$

MCQ উত্তরমালা

01. c	02. a	03. d	04. d	05. d	06. c	07. d	08. a	09. a
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10. $\int \frac{1}{\sqrt{18-2x^2}} dx =$ কত? [BB, JB'22; DB, Ctg.B, CB'21]
 (a) $\frac{1}{\sqrt{2}} \sin^{-1} \frac{x}{3} + c$ (b) $\frac{1}{3\sqrt{2}} \tan^{-1} \frac{x}{3} + c$
 (c) $\frac{1}{6\sqrt{2}} \ln \left| \frac{3+x}{3-x} \right| + c$ (d) $\frac{1}{6\sqrt{2}} \ln \left| \frac{x-3}{x+3} \right| + c$
11. $\int \frac{\tan(\ln x)}{x} dx =$ কত? [CB'21]
 (a) $\ln(\sec^2 x) + c$ (b) $\ln \sec(\ln x) + c$
 (c) $\ln(\sec x) + c$ (d) $\ln(\tan^2 x) + c$
12. $\int e^x (\cos x - \sin x) dx = ?$ [DB'19]
 (a) $e^x \sin x + c$ (b) $e^x \cos x + c$
 (c) $-e^x \cos x + c$ (d) $-e^x \sin x + c$
13. $\int \sqrt{9-x^2} dx =$ কত? [SB'19]
 (a) $\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1} \frac{x}{3} + c$
 (b) $\frac{x}{2} \sqrt{9-x^2} - \frac{9}{2} \sin^{-1} \frac{x}{3} + c$
 (c) $\frac{x}{2} \sqrt{9-x^2} + \frac{3}{2} \sin^{-1} \frac{x}{3} + c$
 (d) $\frac{x}{2} \sqrt{9-x^2} - \frac{3}{2} \sin^{-1} \frac{x}{3} + c$
14. $\int \frac{dx}{\sqrt{36-x^2}} =$ কত? [Ctg.B'17]
 (a) $\sin^{-1} x + C$ (b) $\sin^{-1} \frac{x}{6} + C$
 (c) $\frac{1}{6} \sin^{-1} x + C$ (d) $\frac{1}{6} \sin^{-1} \frac{x}{6} + C$
15. $\int \ln x dx = ?$ [BB'17]
 (a) $\frac{1}{x}$ (b) $x \ln x - x$
 (c) $x \ln x + x$ (d) $\frac{1}{x^2}$
16. যদি $\int f(x) dx = \ln(x + \sqrt{x^2 - 1}) + c$ হয়, তাহলে $f(x) =$ কত?
 (a) $-\frac{1}{\sqrt{x^2-1}}$ (b) $\frac{1}{\sqrt{x^2-1}}$
 (c) $\sqrt{x^2-1}$ (d) $(x^2-1)^{\frac{3}{2}}$
17. $\int e^{5x} \left(5 \ln x + \frac{1}{x} \right) dx = ?$
 (a) $5e^{5x} \ln x + c$ (b) $\frac{e^{5x} \ln x}{5} + c$
 (c) $e^{5x} \ln x + c$ (d) $25e^{5x} \ln x + c$
18. $\int \sqrt{\frac{1-x}{1+x}} dx$ এর মান কত?
 (a) $\sin^{-1} x - \sqrt{1-x^2} + c$ (b) $\sqrt{1-x^2} + c$
 (c) $\sin^{-1} x + c$ (d) $\sin^{-1} x + \sqrt{1-x^2} + c$
19. $\int \frac{dx}{x^2-5x+6} =$ কত?
 (a) $\ln \left| \frac{x-2}{x-3} \right| + c$ (b) $\ln \left| \frac{x+2}{x-3} \right| + c$
 (c) $\ln \left| \frac{x+3}{x-2} \right| + c$ (d) $\ln \left| \frac{x-3}{x-2} \right| + c$
20. $\int \frac{x^2}{x^2-4} dx = ?$
 (a) $-x + \frac{1}{2} \ln \left| \frac{x+2}{x-2} \right| + c$ (b) $x - \ln|x-2| + c$
 (c) $x - \ln \left| \frac{x-2}{x+2} \right| + c$ (d) $x + \ln \left| \frac{x-2}{x+2} \right| + c$
21. $\int \frac{dx}{\sqrt{25-x^2}} = ?$
 (a) $-\cos^{-1}(25x) + c$ (b) $\frac{1}{5} \sin^{-1} x + c$
 (c) $\frac{1}{25} \sin^{-1} \left(\frac{x}{25} \right) + c$ (d) $\sin^{-1} \left(\frac{x}{5} \right) + c$
22. $\int \frac{x+40}{x-40} dx = ?$
 (a) $x - 80 \ln|x+40| + c$
 (b) $\ln \left| \frac{x+40}{x-40} \right| + c$
 (c) $x + 80 \ln|x-40| + c$
 (d) $x - 40 \ln|x-40| + c$

MCQ উত্তরমালা

10. a	11. b	12. b	13. a	14. b	15. b	16. b	17. c	18. d	19. d	20. d	21. d	22. c
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MCQ প্রশ্নের ব্যাখ্যামূলক সমাধান

01. ধরি, $x = a \sec \theta$

$$\therefore dx = a \sec \theta \tan \theta d\theta$$

$$\therefore \int \frac{dx}{\sqrt{x^2-a^2}} = \int \frac{a \sec \theta \tan \theta}{\sqrt{a^2(\sec^2 \theta - 1)}} d\theta = \int \frac{\sec \theta \tan \theta}{\sqrt{\sec^2 \theta - 1}} d\theta$$

$$= \int \frac{\sec \theta \tan \theta}{\tan \theta} d\theta = \int \frac{\sec \theta \tan \theta}{\tan \theta} d\theta = \int \sec \theta d\theta$$

$$= \ln|\sec \theta + \tan \theta| + c_0$$

$$= \ln|\sec \theta + \sqrt{\sec^2 \theta - 1}| + c_0 = \ln \left| \frac{x}{a} + \frac{\sqrt{x^2-a^2}}{a} \right| + c_0$$

$$= \ln|x + \sqrt{x^2-a^2}| + (c_0 - \ln|a|)$$

$$= \ln|x + \sqrt{x^2-a^2}| + c$$

02. $\int e^x (\sin x + \cos x) dx$

$$[\int e^x \{f(x) + f'(x)\} dx = e^x f(x) + c]$$

$$= e^x \sin x + c$$

03.

$$\int_2^3 \frac{2x-1}{x^2-x} dx$$

$$= \int_2^6 \frac{dz}{z} = [\ln z]_2^6$$

$$= \ln 6 - \ln 2$$

$$= \ln \left(\frac{6}{2} \right) = \ln 3$$

04. $\int e^x \left(\frac{1}{x^2} - \frac{1}{x} \right) dx$

$$\text{এখানে, } \frac{d}{dx} \left(-\frac{1}{x} \right) = \frac{1}{x^2}$$

$$\therefore \int e^x \left(\frac{1}{x^2} - \frac{1}{x} \right) dx = e^x \cdot \left(-\frac{1}{x} \right) + c = -\frac{e^x}{x} + c$$

05. $\int e^x \{f(x) + f'(x)\} dx = e^x f(x) + c$

$$\therefore \int e^x (x+1) dx = e^x \cdot x + 1 + c = xe^x + c$$

$$\text{ধরি, } x^2 - x = z$$

$$\therefore (2x-1)dx = dz$$

X	2	3
Z	2	6





06. $\int x \sec^2 x \, dx = x \int \sec^2 x \, dx - \int \left\{ \frac{d}{dx}(x) \int \sec^2 x \, dx \right\} dx$
 $= x \tan x - \int \tan x \, dx = x \tan x - \ln |\sec x| + c$
 $= x \tan x + \ln |\cos x| + c$
07. $\int \frac{x \, dx}{\sqrt{1-x^2}}$; ধরি, $u = 1 - x^2$; $du = -2x \, dx \Rightarrow \frac{du}{-2} = x \, dx$
 $\Rightarrow \int \frac{\frac{du}{-2}}{\sqrt{u}} = -\frac{1}{2} \int u^{-\frac{1}{2}} du = -\frac{1}{2} \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c = -\frac{1}{2} \times 2\sqrt{u} + c$
 $= -\sqrt{u} + c = -\sqrt{1-x^2} + c \therefore f(x) = -\sqrt{1-x^2}$
08. $\int \frac{dx}{9+x^2} = \int \frac{dx}{3^2+x^2} = \frac{1}{3} \tan^{-1} \left(\frac{x}{3} \right) + c$
09. $\int \frac{x \, dx}{\sqrt{1-x}}$ [ধরি, $1-x = z^2 \therefore -dx = 2z \, dz$]
 $= \int \frac{1-z^2}{z} (-2z) \, dz = -2 \int (1-z^2) \, dz$
 $= -2 \left[z - \frac{z^3}{3} \right] + c$
 $= -2z + \frac{2z^3}{3} + c = -2(1-x)^{\frac{1}{2}} + \frac{2}{3}(1-x)^{\frac{3}{2}} + c$
10. $\int \frac{1}{\sqrt{18-2x^2}} \, dx = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{3^2-x^2}} = \frac{1}{\sqrt{2}} \sin^{-1} \frac{x}{3} + c$
11. $\int \frac{\tan(\ln x)}{x} \, dx = \int \tan z \, dz$
 $= \ln |\sec z| + c = \ln |\sec(\ln x)| + c$
 $\left[z = \ln x \Rightarrow dz = \frac{1}{x} \, dx \right]$
12. $\int e^x (\cos x - \sin x) \, dx = \int e^x \left[\cos x + \frac{d}{dx}(\cos x) \right] dx$
 $= e^x \cos x + c$
14. $\int \frac{dx}{\sqrt{36-x^2}} = \int \frac{dx}{\sqrt{6^2-x^2}} = \sin^{-1} \frac{x}{6} + c$

15. $I = \int \ln x \, dx = \ln x \int dx - \int \left\{ \frac{d}{dx}(\ln x) \cdot \int dx \right\} dx$
 $= \ln x \int dx - \int \frac{1}{x} \cdot x \, dx$
 $\ln x \int dx - \int dx = x \ln x - x + c$
16. $\int \frac{\tan(\ln x)}{x} \, dx$
 $= \int \tan z \, dz$ Let, $\ln x = z$
 $= \ln |\sec z| + c \therefore \frac{1}{x} \, dx = dz$
 $= \ln |\sec(\ln x)| + c$
17. $\int e^{5x} \left\{ 5 \ln x + \frac{d}{dx}(\ln x) \right\} \cdot dx$
 $= \int d(\ln x \times e^{5x}) = e^{5x} \ln x + c$
18. $\int \sqrt{\frac{1-x}{1+x}} \, dx = \int \frac{1-x}{\sqrt{1-x^2}} \, dx = \int \frac{1}{\sqrt{1-x^2}} \, dx - \int \frac{x}{\sqrt{1-x^2}} \, dx$
 $= \sin^{-1} x + \sqrt{1-x^2} + c$
19. $\frac{1}{x^2-5x+6} = \frac{1}{(x-3)(x-2)} = -\frac{1}{(x-2)} + \frac{1}{(x-3)}$
 $\int \frac{dx}{x^2-5x+6} = \int \frac{1}{x-3} \cdot dx - \int \frac{1}{x-2} \cdot dx$
 $= \ln|x-3| - \ln|x-2| + c = \ln \left| \frac{x-3}{x-2} \right| + c$
20. $\int \frac{x^2}{x^2-4} \, dx = \int \frac{x^2-4+4}{x^2-4} \, dx = \int dx + 4 \cdot \int \frac{dx}{x^2-4}$
 $= x + \frac{4}{2 \times 2} \ln \left| \frac{x-2}{x+2} \right| + c = x + \ln \left| \frac{x-2}{x+2} \right| + c$
21. $\int \frac{dx}{\sqrt{25-x^2}} = \int \frac{dx}{\sqrt{5^2-x^2}} = \sin^{-1} \left(\frac{x}{5} \right) + c$
22. $\int \frac{x+40}{x-40} \, dx = \int \frac{x-40+80}{x-40} \, dx = \int dx + \int \frac{80}{x-40} \, dx$
 $= x + 80 \ln|x-40| + c$

◆ টাইপভিত্তিক বিগত ৭ বছরের (২০১৯-২০২৫) সৃজনশীল (ক, খ, গ) প্রশ্নের আনালাইসিস:

গুরুত্ব	টাইপের নাম	যতবার প্রশ্ন এসেছে			DB	RB	Ctg.B	BB	SB	JB	CB	Din.B	MB	Mad.B	মোট
		(ক)	(খ)	(গ)											
☆☆	T-09: $\int \frac{dx}{ax^2+bx+c}$, $\int \frac{dx}{\sqrt{ax^2+bx+c}}$ $\int \sqrt{ax^2+bx+c} \, dx$ আকারের যেখানে a, b, c যেকোনো ধ্রুবসংখ্যা	6	3	3	2	2	1	2	2	-	-	-	2	1	12
☆	T-10: $\int \frac{px+q}{ax^2+bx+c} \, dx$, $\int \frac{px+q}{\sqrt{ax^2+bx+c}} \, dx$, $\int (px+q)\sqrt{ax^2+bx+c} \, dx$ আকারের; যেখানে a, b, c, p, q যেকোনো ধ্রুবক	-	1	-	-	-	-	1	-	-	-	-	-	-	1
☆	T-11: $\int \frac{dx}{(ax+b)\sqrt{cx+d}}$; $\int (ax+b)\sqrt{cx+d} \, dx$; $\int \frac{ax+b}{\sqrt{cx+d}} \, dx$ আকারের	-	1	1	-	-	1	1	-	-	-	-	-	-	2
☆☆☆	T-12: অংশক্রমে সমাকলন বা $\int uv \, dx$ সংক্রান্ত	11	6	5	4	2	3	1	1	2	2	2	4	1	22
☆	T-13: অংশক্রমে সমাকলনের ক্ষেত্রে: $\int e^{ax} \{af(x) + f'(x)\} \, dx$ আকৃতির	-	1	-	-	-	-	1	-	-	-	-	-	-	1
☆☆☆	T-14: আংশিক ভগ্নাংশের সাহায্যে সমাকলন	-	16	1	1	2	1	2	2	2	3	1	2	1	18





CQ প্রশ্ন

01. (ক) $\int x e^{-x} dx$ নির্ণয় কর। [RB'25]
 02. $p = 4 - x^2$ [RB'25; Ctg.B'21]
 (খ) $\int \frac{4-p}{p} dx$ নির্ণয় কর।
 03. $f(x) = x + 1, g(x) = e^x$ [BB'25]
 (খ) $\int \frac{f(x-1)g(x)}{\{f(x)\}^2} dx$ নির্ণয় কর।
 04. $f(x) = x + 1$ [BB'25]
 (গ) মান নির্ণয় কর: $\int_{-2}^2 \{f(x-1)^2\} \sqrt{16 - \{f(x-1)^2\}} dx$
 05. $g(x) = \frac{x}{(x-1)(x^2+1)}$ [CB, Mad.B'25; JB'23, 19; MB'21; BB'22; RB, Ctg.B, BB, CB'18]
 (খ) $\int g(x) dx$ নির্ণয় কর।
 06. $f(x) = \cos x$ [Mad.B'25; DB'22, 17; CB'17]
 (গ) $\int e^x f(x) dx$ এর মান নির্ণয় কর।
 07. (ক) $\int \frac{1}{a^2+4x^2} dx$ নির্ণয় কর। [DB'23]
 08. $g(x) = e^x$. [SB'23]
 (খ) $\int \frac{g(2x)}{\{g(x)-1\}\{g(2x)+1\}} dx$ নির্ণয় কর।
 09. (ক) $\int x \ln x dx$ নির্ণয় কর। [JB'23]
 10. $g(x) = x + 2$; $h(x) = (1-x)(x^2+4)$. [JB'23; MB'21; JB'19]
 (গ) $\int \frac{g(x)}{h(x)} dx$ নির্ণয় কর।
 11. দৃশ্যকল্প-১: $f(x) = \sin x$; দৃশ্যকল্প-২: $g(x) = e^x$. [DB'22]
 (খ) $\int g(x)f\left(\frac{\pi}{2} - x\right) dx$ এর যোগজ নির্ণয় কর।
 12. $g(x) = \frac{x^2}{x^2-4}$ [Ctg.B'21]
 (গ) $\int g(x) dx$ নির্ণয় কর।

CQ প্রশ্নের সমাধান

01. (ক) Solⁿ: $\int x e^{-x} dx = -x e^{-x} - \int \left[\frac{d}{dx}(x) \cdot \int e^{-x} dx \right] dx$
 $= -x e^{-x} + \int 1 \cdot e^{-x} dx = -x e^{-x} - e^{-x} + c$
 $= -e^{-x}[1+x] + c$ (Ans.)
 02. (খ) Solⁿ: দেওয়া আছে, $p = 4 - x^2$
 $\therefore \int \frac{4-p}{p} dx = \int \frac{4-(4-x^2)}{4-x^2} dx = 4 \int \frac{dx}{2^2-x^2} - \int \frac{4-x^2}{4-x^2} dx$
 $= 4 \times \frac{1}{2 \times 2} \ln \left| \frac{2+x}{2-x} \right| - \int dx = \ln \left| \frac{2+x}{2-x} \right| - x + c$ (Ans.)
 03. (খ) Solⁿ: দেওয়া আছে, $f(x) = x + 1, g(x) = e^x$;
 $\therefore \int \frac{f(x-1)g(x)}{\{f(x)\}^2} dx = \int \frac{(x-1+1)e^x}{(x+1)^2} dx$
 $= \int e^x \left\{ \frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} \right\} dx = \int e^x \left\{ \frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} \right\} dx$
 $= \int e^x \left\{ \frac{1}{x+1} + \frac{d}{dx} \left(\frac{1}{x+1} \right) \right\} dx$
 $\left[\because \frac{d}{dx} \left(\frac{1}{x+1} \right) = -(x+1)^{-2} = -\frac{1}{(x+1)^2} \right]$
 $= \frac{e^x}{1+x} + c$ [$\because e^x \{f(x) + f'(x)\} = e^x f(x) + c$]
 $\therefore \int \frac{f(x-1)g(x)}{\{f(x)\}^2} = \frac{e^x}{1+x} + c$ (Ans.)
 04. (গ) Solⁿ: দেওয়া আছে, $f(x) = x + 1 \dots \dots \dots (i)$
 $\therefore \int_{-2}^2 \{f(x-1)\}^2 \sqrt{16 - \{f(x-1)\}^2} dx$
 $= \int_{-2}^2 (x-1+1)^2 \sqrt{16 - (x-1+1)^2} dx$
 $= \int_{-2}^2 x^2 \sqrt{16 - x^2} dx$
 ধরি, $x = 4 \sin \theta$
 $\Rightarrow dx = 4 \cos \theta d\theta$

x	-2	2
θ	$-\frac{\pi}{6}$	$\frac{\pi}{6}$

$$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (4 \sin \theta)^2 \sqrt{16 - (4 \sin \theta)^2} \cdot 4 \cos \theta d\theta$$

$$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 16 \sin^2 \theta \cdot 4 \sqrt{1 - \sin^2 \theta} \cdot 4 \cos \theta d\theta$$

$$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 16 \sin^2 \theta \cdot 4 \cos \theta \cdot 4 \cos \theta d\theta$$

$$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 64 (2 \sin \theta \cos \theta)^2 d\theta = 64 \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \sin^2 2\theta d\theta$$

$$= 32 \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (1 - \cos 4\theta) d\theta = 32 \left[\theta \right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} - 8 \left[\sin 4\theta \right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$$

$$= 32 \left(\frac{\pi}{6} + \frac{\pi}{6} \right) - 8 \left[\sin \frac{2\pi}{3} - \sin \left(-\frac{2\pi}{3} \right) \right]$$

$$= 32 \times \frac{2\pi}{6} - 8 \times 2 \sin \frac{2\pi}{3}$$

$$= \frac{32}{3} \pi - 8 \times 2 \times \frac{\sqrt{3}}{2} = \frac{32}{3} \pi - 8\sqrt{3}$$
 (Ans.)
 05. (খ) Solⁿ: দেওয়া আছে, $g(x) = \frac{x}{(x-1)(x^2+1)}$
 $\therefore \int g(x) dx = \int \frac{x}{(x-1)(x^2+1)} dx$
 ধরি, $\frac{x}{(x-1)(x^2+1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$
 $\Rightarrow x = A(x^2+1) + (Bx+C)(x-1)$
 $\Rightarrow x = (A+B)x^2 + (C-B)x + (A-C)$
 $x = 1$ বসিয়ে পাই,
 $1 = A(1+1) + (B \cdot 1 + C)(1-1) \Rightarrow A = \frac{1}{2}$
 x^2 এর সহগ সমীকৃত করে পাই, $0 = A+B \Rightarrow B = -A = -\frac{1}{2}$
 x এর সহগ সমীকৃত করে পাই, $1 = C-B \Rightarrow C = 1+B = \frac{1}{2}$
 $\therefore \frac{x}{(x-1)(x^2+1)} = \frac{\frac{1}{2}}{x-1} + \frac{-\frac{1}{2}x + \frac{1}{2}}{x^2+1} = \frac{1}{2} \left[\frac{1}{x-1} - \frac{x-1}{x^2+1} \right]$





$$\begin{aligned} \therefore \int \frac{x}{(x-1)(x^2+1)} dx &= \int \frac{1}{2} \left[\frac{1}{x-1} - \frac{x-1}{x^2+1} \right] dx \\ &= \frac{1}{2} \int \left[\frac{1}{x-1} - \frac{x}{x^2+1} + \frac{1}{x^2+1} \right] dx \\ &= \frac{1}{2} \left[\int \frac{dx}{x-1} - \frac{1}{2} \int \frac{2x dx}{x^2+1} + \int \frac{dx}{1+x^2} \right] \\ &= \frac{1}{2} \left[\ln|x-1| - \frac{1}{2} \ln|x^2+1| + \tan^{-1} x \right] + c \text{ (Ans.)} \end{aligned}$$

06.

(গ) Solⁿ: দেওয়া আছে, $f(x) = \cos x$

$$\therefore \int e^x f(x) dx = \int e^x \cos x dx$$

$$\text{ধরি, } I = \int e^x \cos x dx$$

$$= e^x \int \cos x dx - \int \left\{ \frac{d}{dx} (e^x) \int \cos x dx \right\} dx$$

$$= e^x \sin x - \int e^x \sin x dx$$

$$= e^x \sin x - \left[e^x \int \sin x dx - \int \left\{ \frac{d}{dx} (e^x) \int \sin x dx \right\} dx \right]$$

$$= e^x \sin x - [e^x(-\cos x) - \int e^x(-\cos x) dx]$$

$$= e^x \sin x - [-e^x \cos x + \int e^x \cos x dx]$$

$$I = e^x \sin x - [-e^x \cos x + I] \quad [\because \int e^x \cos x dx = I]$$

$$\Rightarrow I = e^x \sin x + e^x \cos x - I$$

$$\therefore 2I = e^x(\sin x + \cos x) \Rightarrow I = \frac{e^x}{2}(\sin x + \cos x)$$

07.

$$\begin{aligned} \text{(ক) Solⁿ: } \int \frac{1}{a^2+4x^2} dx &= \frac{1}{4} \int \frac{1}{\frac{a^2}{4}+x^2} dx = \frac{1}{4} \int \frac{dx}{\left(\frac{a}{2}\right)^2+x^2} \\ &= \frac{1}{4} \times \frac{2}{a} \tan^{-1} \frac{2x}{a} + c = \frac{1}{2a} \tan^{-1} \frac{2x}{a} + c \text{ (Ans.)} \end{aligned}$$

08.

(খ) Solⁿ: দেওয়া আছে, $g(x) = e^x$

$$\int \frac{g(2x)}{\{g(x)-1\}\{g(2x)+1\}} dx = \int \frac{e^{2x} dx}{(e^x-1)(e^{2x}+1)} \quad \left| \begin{array}{l} \text{ধরি, } e^x = y \\ \Rightarrow e^x dx = dy \end{array} \right.$$

$$= \int \frac{y^2 dy}{(y-1)(y^2+1)}$$

$$= \int \frac{y-1}{(y-1)(y^2+1)} dy + \int \frac{dy}{(y-1)(y^2+1)}$$

$$= \int \frac{dy}{y^2+1} + \int \frac{dy}{(y-1)(y^2+1)} = \tan^{-1} y + \int \frac{dy}{(y-1)(y^2+1)} + c_1$$

$$\text{এখন, } \frac{1}{(y-1)(y^2+1)} \equiv \frac{A}{y-1} + \frac{By+C}{y^2+1}$$

$$\Rightarrow 1 \equiv A(y^2+1) + (By+C)(y-1)$$

$$\Rightarrow 1 \equiv Ay^2 + A + By^2 - By + Cy - C$$

$$\Rightarrow 1 \equiv (A+B)y^2 + (-B+C)y + A-C$$

$$y^2, y, \text{ ধ্রুবক পদ সমীকৃত করে পাই, } A+B=0 \dots \dots \dots (i);$$

$$-B+C=0 \dots \dots \dots (ii); A-C=1 \dots \dots \dots (iii)$$

$$(i) \text{ হতে পাই, } A = -B;$$

$$(ii) \text{ নং হতে পাই, } -B+C=0 \Rightarrow A+C=0 \dots \dots \dots (iv)$$

$$(iii) \text{ ও } (iv) \text{ সমাধান করে পাই, } 2A=1 \therefore A=\frac{1}{2} \quad B=-\frac{1}{2}$$

$$(iv) \text{ হতে পাই, } \frac{1}{2} + C = 0 \therefore C = -\frac{1}{2}$$

$$\therefore \frac{1}{(y-1)(y^2+1)} = \frac{1}{2(y-1)} - \frac{y+1}{2(y^2+1)}$$

$$\therefore \int \frac{y dy}{(y-1)(y^2+1)} = \tan^{-1} y + \int \frac{dy}{2(y-1)} - \int \frac{y+1}{2(y^2+1)} dy + c_1$$

$$= \tan^{-1} y + \frac{1}{2} \ln|y-1| - \int \frac{y dy}{2(y^2+1)} - \int \frac{1}{2(y^2+1)} + c_2$$

$$= \tan^{-1} y + \frac{1}{2} \ln|y-1| - \frac{1}{4} \ln|y^2+1| - \frac{1}{2} \tan^{-1} y + c$$

$$= \frac{1}{2} \tan^{-1} y + \frac{1}{2} \ln|y-1| - \frac{1}{4} \ln|y^2+1| + c \text{ (Ans.)}$$

09.

$$\begin{aligned} \text{(ক) Solⁿ: } \int x \ln x dx &= \ln x \int x dx - \int \frac{d}{dx} (\ln x) \int x dx dx \\ &= \frac{x^2 \ln x}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx = \frac{x^2 \ln x}{2} - \int \frac{x}{2} dx \\ &= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + c = \frac{x^2}{4} (2 \ln x - 1) + c \text{ (Ans.)} \end{aligned}$$

10.

(গ) Solⁿ: $g(x) = x+2$; $h(x) = (1-x)(x^2+4)$

$$\therefore \int \frac{g(x)}{h(x)} dx = \int \frac{(x+2)}{(1-x)(x^2+4)} dx$$

$$\text{ধরি, } \frac{x+2}{(1-x)(x^2+4)} \equiv \frac{A}{1-x} + \frac{Bx+C}{x^2+4}$$

$$\Rightarrow (x+2) \equiv A(x^2+4) + (Bx+C)(1-x) \dots \dots \dots (i)$$

$$\therefore x=1 \text{ বসিয়ে পাই, } 3=5A \Rightarrow A=\frac{3}{5}$$

$$\text{এখন, } x+2 \equiv Ax^2+4A-Bx^2-Cx+Bx+C$$

$$\Rightarrow x+2 \equiv x^2(A-B) + x(B-C) + (4A+C)$$

$$\therefore x^2 \text{ এর সহগ সমীকৃত করে পাই, } 0=A-B$$

$$\Rightarrow B=A=\frac{3}{5}$$

$$\therefore x \text{ এর সহগ সমীকৃত করে পাই,}$$

$$1=B-C \Rightarrow C=B-1=\frac{3}{5}-1=-\frac{2}{5}$$

$$\therefore I = \int \frac{g(x)}{h(x)} dx = \int \frac{3}{5(1-x)} dx + \int \frac{\frac{3}{5}x-\frac{2}{5}}{x^2+4} dx$$

$$= \frac{3}{5} \int \frac{dx}{1-x} + \frac{1}{5} \int \frac{3x-2}{x^2+4} dx$$

$$= \frac{-3}{5} \ln|1-x| + \frac{1}{5} \left[\frac{3}{2} \int \frac{2x}{x^2+4} dx - \int \frac{2}{x^2+2^2} dx \right] + c_1$$

$$= -\frac{3}{5} \ln|1-x| + \frac{1}{5} \cdot \frac{3}{2} \ln|x^2+4| - 2 \cdot \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$$

$$= -\frac{3}{5} \ln|1-x| + \frac{3}{10} \ln|x^2+4| - \tan^{-1} \left(\frac{x}{2} \right) + c \text{ (Ans.)}$$

11.

(খ) Solⁿ: ধরি, $I = \int g(x) f\left(\frac{\pi}{2} - x\right) dx$

$$= \int e^x \cdot \sin\left(\frac{\pi}{2} - x\right) dx = \int e^x \cos x dx$$

$$= \cos x \int e^x dx - \int \left[\frac{d}{dx} (\cos x) \int e^x dx \right] dx$$

$$= \cos x \cdot e^x - \int -\sin x \cdot e^x dx$$

$$= e^x \cos x + \int \sin x \cdot e^x dx$$

$$= e^x \cos x + \sin x \int e^x dx - \int \left[\frac{d}{dx} (\sin x) \cdot \int e^x dx \right] dx$$

$$\text{সুতরাং, } I = e^x \cos x + e^x \sin x - \int \cos x \cdot e^x dx + c_1$$

$$\Rightarrow I = e^x(\cos x + \sin x) - I + c_1$$

$$\Rightarrow 2I = e^x(\sin x + \cos x) + c_1$$

$$\Rightarrow I = \frac{e^x}{2}(\sin x + \cos x) + \frac{c_1}{2}$$

$$\therefore I = \frac{e^x}{2}(\sin x + \cos x) + c \left[\text{যেখানে, } c = \frac{c_1}{2} \right] \text{ (Ans.)}$$

12.

$$\text{(গ) Solⁿ: } g(x) = \frac{x^2}{x^2-4} = \frac{x^2-4+4}{x^2-4} = 1 - \frac{4}{4-x^2}$$

$$\therefore \int g(x) dx = \int \left(1 - \frac{4}{4-x^2} \right) dx = \int dx - 4 \int \frac{1}{2^2-x^2} dx$$

$$= x - 4 \times \frac{1}{2 \cdot 2} \ln \left| \frac{2-x}{2+x} \right| + c = x - \ln \left| \frac{2+x}{2-x} \right| + c$$





লেকচার-২৩

♦ টাইপভিত্তিক বিগত ৭ বছরের (২০১৯-২০২৫) MCQ প্রশ্নের অ্যানালাইসিস:

গুরুত্ব	টাইপের নাম	DB	RB	Ctg.B	BB	SB	JB	CB	Din.B	MB	Mad.B	মোট
☆	T-15: নির্দিষ্ট যোগজ সম্পর্কিত মূল উপপাদ্য এর প্রয়োগ ও সাধারণ সমস্যা	-	-	-	1	1	-	-	-	-	-	2
☆☆	T-16: নির্দিষ্ট যোগজ ব্যবহার করে ক্ষেত্রফল নির্ণয় সংক্রান্ত	1	1	3	2	6	1	3	1	-	-	19
☆☆☆	T-17: বিবিধ	2	2	1	3	3	4	4	1	5	-	25

MCQ প্রশ্ন

01. $\int_0^a \sqrt{a^2 - x^2} dx =$ কত?

- (a) $-\frac{a^2\pi}{2}$ (b) $\frac{a^2\pi}{2}$ (c) $-\frac{a^2\pi}{4}$ (d) $\frac{a^2\pi}{4}$ [RB'25]

02. $\int_1^e \frac{dx}{x(1+\ln x)} = ?$

- (a) $\ln e - 1$ (b) $\ln 2$ (c) e (d) $e + 1$ [JB'25; DB, Ctg.B'22; All.B'18]

03. $\int_0^{\frac{\pi}{2}} \sqrt{1 + \cos x} dx$ এর মান কত?

- (a) -1 (b) $+1$ (c) -2 (d) $+2$ [CB'25]

04. $\int_0^1 \frac{1}{1+2x} dx$ এর মান কত?

- (a) $\frac{1}{2} \ln \frac{1}{3}$ (b) $\frac{1}{2} \ln 3$ (c) $\ln \frac{1}{3}$ (d) $\ln 3$ [MB'25]

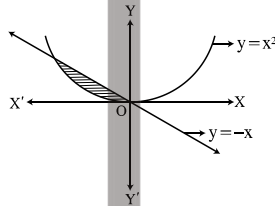
05. $\int_0^{\pi/2} \sqrt{1 + \sin 2x} dx = ?$

- (a) -2 (b) 0 (c) 1 (d) 2 [DB'23]

06. $\int_0^{\frac{\pi}{2}} \cos x dx = ?$

- (a) 0 (b) 1 (c) -1 (d) ∞ [RB'23]

07.



চিত্রের ছায়াঘেরা অংশের ক্ষেত্রফল কত বর্গ একক?

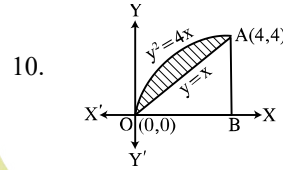
- (a) $-\frac{5}{6}$ (b) $-\frac{1}{6}$ (c) $\frac{1}{6}$ (d) $\frac{5}{6}$ [Ctg.B'23]

08. $\int_0^1 \frac{x^2}{x^3+2} dx =$ কত?

- (a) $\frac{1}{3} \ln \left(\frac{3}{2}\right)$ (b) $\ln \left(\frac{3}{2}\right)$ (c) $\frac{1}{3} \ln \left(\frac{2}{3}\right)$ (d) $\ln \left(\frac{2}{3}\right)$ [Ctg.B'23]

09. $\int_0^{1-x} \frac{1-x}{1+x} dx$ এর মান কোনটি?

- (a) $2 \ln 2 + 1$ (b) $2 \ln 2 - 1$ (c) $\ln 2 + 1$ (d) $\ln 2 - 2$ [CB'23; MB'21]



উপর্যুক্ত চিত্রের ছায়াঘেরা অংশের ক্ষেত্রফল কত বর্গ একক?

- (a) $\frac{16}{3}$ (b) $\frac{4}{3}$ (c) $\frac{8}{3}$ (d) $\frac{32}{3}$ [Din.B'23]

11. $I = \int_1^e \frac{dx}{x(1+\ln x)}$ হলে I এর মান কত?

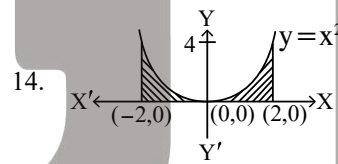
- (a) e (b) $e + 1$ (c) $\ln e - 1$ (d) $\ln 2$ [DB, Ctg.B'22; All.B'18]

12. $\int_0^3 f(x) dx = 4$ হলে, $\int_2^5 f(x-2) dx =$ কত?

- (a) 0 (b) 4 (c) 3 (d) 2 [SB'22]

13. $\frac{x^2}{16} + \frac{y^2}{25} = 1$ উপবৃত্তটির ক্ষেত্রফল কত বর্গ একক?

- (a) 16π (b) 20π (c) 25π (d) 400π [DB, SB'21; SB, Din.B'17]



চিত্রে ছায়াঘেরা অংশের ক্ষেত্রফল কত বর্গ একক?

- (a) $\frac{32}{3}$ (b) $\frac{16}{3}$ (c) $\frac{8}{3}$ (d) $\frac{4}{3}$ [SB'21; Din.B'19]

15. $y^2 = 4ax$ ও $x^2 = 4ay$ পরাবৃত্ত দুটি দ্বারা সীমাবদ্ধ সমতল ক্ষেত্রের ক্ষেত্রফল নির্ণয়ের জন্য কোন যোগজটি সঠিক?

- (a) $\int_0^{4a} \left(2\sqrt{ax} - \frac{x^2}{4a}\right) dx$ (b) $\int_0^{4a} \left(2\sqrt{ax} - \frac{4a}{x^2}\right) dx$ (c) $\int_0^{4a} \left(\frac{4a}{x^2} - 2\sqrt{ax}\right) dx$ (d) $\int_0^{4a} \left(\sqrt{ax} - \frac{x^2}{2a}\right) dx$ [BB'21]

MCQ উত্তরমালা

01. d	02. b	03. d	04. b	05. d	06. b	07. c	08. a	09. b	10. c	11. d	12. b	13. b	14. b	15. a
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16. $\int_0^1 \frac{1-x}{1+x} dx = ?$ [MB'21]
 (a) $\ln 2 - 2$ (b) $1 - 2 \ln 2$
 (c) $2 \ln 2 - 1$ (d) $2 \ln 2 + 2$
17. $y^2 = x, y = 0, x = 1$ এবং $x = 4$ দ্বারা আবদ্ধ ক্ষেত্রের ক্ষেত্রফল কত বর্গ একক?
 (a) $\frac{27}{2}$ (b) $\frac{21}{2}$ (c) $\frac{14}{3}$ (d) $\frac{11}{2}$
18. $\int_2^3 \frac{2x-1}{x(x-1)} dx$ এর মান নিচের কোনটি?
 (a) $\ln 4$ (b) $2 \ln 3$ (c) $\ln 3$ (d) $\ln 2$
19. $\int_0^{\pi/2} \sin^2 2x dx = ?$
 (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{5\pi}{6}$

20. নিচের তথ্যগুলো লক্ষ কর-
 (i) $\int_{-4}^{11} |x-1| dx = \int_{-4}^0 (1-x) dx + \int_0^{11} (x-1) dx$
 (ii) $\frac{x^2}{4} + \frac{y^2}{9} = 1$ উপবৃত্তের ক্ষেত্রফল 6π বর্গ একক
 (iii) $\int e^x \{f(x) + f'(x)\} dx = e^x f(x) + c$
 নিচের কোনটি সঠিক?
 (a) i, ii (b) ii, iii (c) i, iii (d) i, ii, iii
21. $y = \sin^2 x$ বক্ররেখার একটি চাপ দ্বারা x অক্ষ দ্বারা আবদ্ধ ক্ষেত্রের ক্ষেত্রফলের অর্ধেক-
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 1 (d) 2
22. $x^2 = 2y$ ও $y^2 = 2x$ পরাবৃত্তদ্বয় দ্বারা আবদ্ধ ক্ষেত্রের ক্ষেত্রফল-
 (a) $\frac{4}{3}$ (b) $\frac{2}{3}$ (c) $\frac{16}{3}$ (d) $\frac{64}{3}$

MCQ উত্তরমালা

16. c

17. c

18. c

19. b

20. b

21. b

22. a

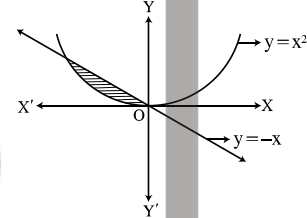
MCQ প্রশ্নের ব্যাখ্যামূলক সমাধান

01. $\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$
 $[y = \sqrt{a^2 - x^2} \Rightarrow y^2 = a^2 - x^2 \Rightarrow x^2 + y^2 = a^2]$
02. $\int_1^e \frac{dx}{x(1+\ln x)}$
 ধরি, $1 + \ln x = z$
 $\Rightarrow \frac{1}{x} dx = dz$

x	1	e
z	1	2

 $= \int_1^2 \frac{dz}{z} = [\ln z]_1^2$
 $= \ln 2 - \ln 1 = \ln 2$
03. $\int_0^{\pi} \sqrt{1 + \cos x} dx = \int_0^{\pi} \sqrt{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}} dx$
 $= \int_0^{\pi} \sqrt{2} \cos \frac{x}{2} dx = \sqrt{2} \int_0^{\pi} \cos \frac{x}{2} dx$
 $= 2\sqrt{2} \left[\sin \frac{x}{2} \right]_0^{\pi} = 2\sqrt{2} (\sin \frac{\pi}{2} - \sin 0)$
 $= 2\sqrt{2} (\frac{1}{\sqrt{2}} - 0) = 2\sqrt{2} \times \frac{1}{\sqrt{2}} = 2$
04. $\int_0^1 \frac{1}{1+2x} dx = \left[\frac{1}{2} \ln |1+2x| \right]_0^1$
 $= \frac{1}{2} [\ln |1+2| - \ln |1+0|] = \frac{1}{2} \ln 3$
05. $\int_0^{\pi} \sqrt{1 + \sin 2x} dx$
 $= \int_0^{\pi} \sqrt{\sin^2 x + 2 \sin x \cos x + \cos^2 x} dx$
 $= \int_0^{\pi} \sqrt{(\sin x + \cos x)^2} dx = \int_0^{\pi} (\sin x + \cos x) dx$
 $= [-\cos x + \sin x]_0^{\pi} = -\cos \frac{\pi}{2} + \sin \frac{\pi}{2} + \cos 0 - \sin 0$
 $= -0 + 1 + 1 - 0 = 2$
06. $\int_0^{\pi} \cos x dx = [\sin x]_0^{\pi}$
 $= \sin \frac{\pi}{2} - \sin 0 = 1 - 0 = 1$

07. দেওয়া আছে, $y = -x$ এবং $y = x^2$



$\Rightarrow -x = x^2 \Rightarrow x(x+1) = 0 \therefore x = 0, -1$
 $x = 0$ হলে $y = 0$; $x = -1$ হলে $y = 1$
 $\therefore$ ক্ষেত্রফল $= \int_0^{-1} (x^2 + x) dx$
 $= \frac{1}{3} [x^3]_0^{-1} + \frac{1}{2} [x^2]_0^{-1} = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$

08. ধরি, $z = x^3 + 2$; $dz = 3x^2 dx$

x	0	1
z	2	3

 $\int_0^1 \frac{x^2}{x^3+2} dx = \frac{1}{3} \int_2^3 \frac{3x^2}{x^3+2} dx = \frac{1}{3} \int_2^3 \frac{dz}{z} = \frac{1}{3} [\ln |z|]_2^3$
 $= \frac{1}{3} [\ln 3 - \ln 2] = \frac{1}{3} \ln \left(\frac{3}{2} \right)$

09. $\int_0^1 \frac{1-x}{1+x} dx = \int_0^1 \left(\frac{-x-1+2}{1+x} \right) dx = \int_0^1 \left(\frac{-(1+x)}{1+x} + \frac{2}{1+x} \right) dx$
 $= [-x + 2 \ln |1+x|]_0^1 = 2 \ln 2 - 1$

10. $\int_0^4 (\sqrt{4x} - x) dx = \int_0^4 (2\sqrt{x} - x) dx = \left[2 \cdot \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} - \frac{x^2}{2} \right]_0^4$
 $= \left[\frac{4}{3} x^{\frac{3}{2}} - \frac{x^2}{2} \right]_0^4 = \frac{4}{3} (4)^{\frac{3}{2}} - \frac{4^2}{2} = \frac{32}{3} - 8 = \frac{32-24}{3} = \frac{8}{3}$

- 11.

$I = \int_1^e \frac{dx}{x(1+\ln x)} = \int_1^2 \frac{dz}{z} = [\ln |z|]_1^2$
 $= \ln |2| - \ln |1|$
 $= \ln 2 - 0 = \ln 2$

ধরি, $1 + \ln x = z$
 $\Rightarrow \frac{1}{x} dx = dz$

x	1	e
z	1	2

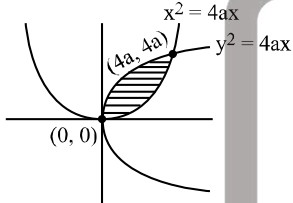




12. $\int_0^3 f(z) dz = 4 \therefore \int_2^5 f(x-2) dx$
 ধরি, $x-2 = z \therefore x = z+2 \therefore dx = dz \therefore \int_0^3 f(z) dz = 4$

13. $\frac{x^2}{16} + \frac{y^2}{25} = 1 \Rightarrow \frac{x^2}{4^2} + \frac{y^2}{5^2} = 1 \therefore a = 4, b = 5$
 $\therefore$ উপবৃত্তের ক্ষেত্রফল $= \pi ab = 4 \times 5\pi = 20\pi$ বর্গ একক

14. নির্ণেয় ক্ষেত্রফল $= 2 \int_0^2 x^2 dx$
 $= \frac{2}{3} [x^3]_0^2 = \frac{2}{3} [2^3 - 0] = \frac{16}{3}$ বর্গ একক



15.

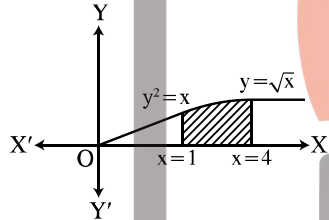
$x^2 = 4ay$ এবং $y^2 = 4ax$ সমাধান করে পাই, ছেদবিন্দুগুলো $(0,0), (4a, 4a)$

$y^2 = 4ax \Rightarrow y = 2\sqrt{ax}$; আবার, $x^2 = 4ay \Rightarrow y = \frac{x^2}{4a}$

$\therefore$ নির্ণেয় ক্ষেত্রফল $= \int_0^{4a} \left(2\sqrt{ax} - \frac{x^2}{4a} \right) dx$ বর্গ একক।

16. $\int_0^1 \frac{1-x}{1+x} dx = \int_0^1 \frac{2-(1+x)}{1+x} dx = \int_0^1 \left(\frac{2}{1+x} - 1 \right) dx$
 $= [2 \ln|1+x| - x]_0^1 = 2 \ln 2 - 1$

17. $y^2 = x, y = 0, x = 1, x = 4$



$\therefore$ নির্ণেয় ক্ষেত্রফল $= \int_1^4 y dx$ বর্গ একক

$= \int_1^4 \sqrt{x} dx = \int_1^4 x^{\frac{1}{2}} dx$

$= \frac{2}{3} [x^{\frac{3}{2}}]_1^4 = \frac{2}{3} (8 - 1) = \frac{14}{3}$ বর্গ একক

18. $\int_2^3 \frac{2x-1}{x^2-x} dx = \int_{x=2}^{x=3} \frac{d(x^2-x)}{x^2-x} dx$
 $= [\ln|x^2-x|]_2^3 = \ln|9-3| - \ln|4-2|$
 $= \ln\left(\frac{9-3}{4-2}\right) = \ln\left(\frac{6}{2}\right) = \ln 3$

19. $\int_0^{\pi/2} \sin^2 2x dx = \int_0^{\pi/2} \frac{1}{2} (1 - \cos 4x) dx$
 $= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/2} = \frac{1}{2} \left(\frac{\pi}{2} - \frac{\sin 2\pi}{4} - 0 - \frac{\sin 0}{4} \right) = \frac{\pi}{4}$

20. (i) $|x-1| = \begin{cases} +(x-1) \text{ hLb, } x-1 > 0 \text{ ev } x > 1 \\ -(x-1) \text{ hLb, } x-1 < 0 \text{ ev } x < 1 \end{cases}$
 $\therefore \int_{-4}^{11} |x-1| dx = \int_{-4}^1 (1-x) dx + \int_1^{11} (x-1) dx$

(ii) $\frac{x^2}{4} + \frac{y^2}{9} = 1$ বা, $\frac{x^2}{2^2} + \frac{y^2}{3^2} = 1$

উপবৃত্ত, $a = 2, b = 3$

$\therefore \text{Area} = \pi ab = \pi \times 2 \times 3 = 6\pi$

(iii) $\frac{d}{dx} [e^x f(x) + c] = \frac{d}{dx} [e^x]$

$f(x) + e^x \frac{d}{dx} [f(x)] + 0$

$= e^x f(x) + e^x f'(x) = e^x [f(x) + f'(x)]$

$\therefore \int e^x \{f(x) + f'(x)\} dx = e^x f(x) + c$

21. $A = \int_0^{\pi/2} \sin^2 x dx = \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2x) dx$
 $= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/2} = \frac{\pi}{4}$

22. $a = \frac{1}{2} \therefore$ ক্ষেত্রফল $= \frac{16}{3} a^2 = \frac{16}{3} \left(\frac{1}{2} \right)^2 = \frac{4}{3}$

♦ টাইপভিত্তিক বিগত ৭ বছরের (২০১৯-২০২৫) সৃজনশীল (ক, খ, গ) প্রশ্নের অ্যানালাইসিস:

গুরুত্ব	টাইপের নাম	যতবার প্রশ্ন এসেছে			DB	RB	Ctg.B	BB	SB	JB	CB	Din.B	MB	Mad.B	মোট
		(ক)	(খ)	(গ)											
***	T-15: নির্দিষ্ট যোগজ সম্পর্কিত মূল উপপাদ্য এর প্রয়োগ ও সাধারণ সমস্যা	12	25	8	3	7	4	4	5	7	4	7	4	-	45
***	T-16: নির্দিষ্ট যোগজ ব্যবহার করে ক্ষেত্রফল নির্ণয় সংক্রান্ত	1		37	5	3	6	5	2	5	4	4	4	-	38
*	T-17: বিবিধ		2	1	1	-	1	-	-	-	-	1	-	-	3

CQ প্রশ্ন

01. দৃশ্যকল্প-২: $y^2 = 16x$

[DB'25]

(গ) দৃশ্যকল্প-২: এর পরাবৃত্ত এবং এর উপকেন্দ্রিক লম্ব দ্বারা

সীমাবদ্ধ ক্ষেত্রের ক্ষেত্রফল নির্ণয় কর।

02. $p = 4 - x^2$

[RB'25; JB'21]

(গ) $\int_{-1}^1 (4-p)\sqrt{p} dx$ এর মান নির্ণয় কর।

03. $y^2 = x \dots \dots \dots$ (i) ও $y - x + 2 = 0 \dots \dots \dots$ (ii)

দুইটি সমীকরণ।

[Ctg.B, JB, MB'25; BB'22]

(গ) (i) ও (ii) নং রেখা দ্বারা আবদ্ধ ক্ষেত্রের ক্ষেত্রফল নির্ণয় কর।

(চিত্র আবশ্যিক)

04. দৃশ্যপট-১: $h(x) = \sin x$

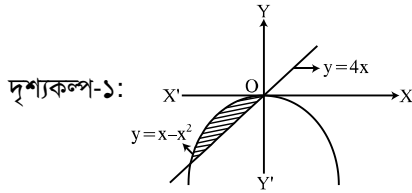
[SB'25; CB'22]

(খ) $\int_0^{\pi/4} \frac{\left\{ h\left(\frac{\pi}{2}-2x\right) \right\}^2}{\left\{ h\left(\frac{\pi}{2}-2x\right) \right\}^2 - \{h(2x)\}^2} dx$ এর মান নির্ণয় কর।





05.



দৃশ্যকল্প-১: এ বর্ণিত চিত্রের ছায়াবৃত্ত অংশের ক্ষেত্রফল নির্ণয় কর।

[RB'23]

06.

$f(x) = \sin x$
(খ) $\int_0^{\frac{\pi}{2}} \frac{f'(x)}{[f(x)]^2 - 16} \{f(x) - 3\} dx$ নির্ণয় কর।

[Ctg.B'23]

07.

দৃশ্যকল্প-২: $x^2 + y^2 = 64$; $y = 5$.
(গ) দৃশ্যকল্প-২ এর বৃত্ত ও সরলরেখা দ্বারা আবদ্ধ ক্ষুদ্রতর অংশের ক্ষেত্রফল নির্ণয় কর।

[JB'23]

08.

(ক) $\int_1^2 \ln 2x dx$ এর মান নির্ণয় কর।

[BB'22]

09.

$P(x) = \cos x$
(ক) $\int_0^{\frac{\pi}{4}} \frac{dx}{1 + \cos 2x}$ এর মান নির্ণয় কর।
(খ) $\int_0^{\frac{\pi}{2}} x^2 P(x) dx$ এর মান নির্ণয় কর।

[Ctg.B'21]

10.

$g(x) = y^2 - 8x$ এবং $h(x) = x^2 - 8y$.
(গ) $g(x) = 0$ এবং $h(x) = 0$ বক্ররেখা দ্বারা আবদ্ধ ক্ষেত্রের ক্ষেত্রফল নির্ণয় কর।

[JB'21]

11.

দৃশ্যকল্প-২: $2x^2 + 2y^2 = 50$
(গ) যোগজীকরণের সাহায্যে দৃশ্যকল্প-২ এবং $x + y = 5$ রেখা দ্বারা আবদ্ধ ক্ষুদ্রতর অংশের ক্ষেত্রফল নির্ণয় কর।

12.

দৃশ্যকল্প-২: $x^2 + y^2 = 1$ ও $y^2 = 1 - x$ দুইটি বক্ররেখা।
(গ) দৃশ্যকল্প-২ হতে প্রমাণ কর যে, বক্ররেখা দুইটির সীমাবদ্ধ ক্ষুদ্রতর অংশের ক্ষেত্রফল $\left(\frac{\pi}{2} - \frac{4}{3}\right)$ বর্গএকক।

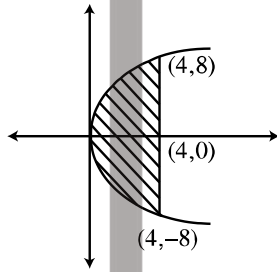
13.

$l(x) = \sin 2x$ and $g(x) = \log x$
(ক) $\int l\left(\frac{x^0}{2}\right) dx$ নির্ণয় কর।
(খ) $\int \frac{\cos 2x}{(\sqrt{3+l(x)})^3} dx$ নির্ণয় কর।
(গ) $\int \frac{1}{x[1+g(x)]^3} dx$ নির্ণয় কর।

CQ প্রশ্নের সমাধান

01.

(গ) Solⁿ: দৃশ্যকল্প-২ হতে পাই, $y^2 = 16x \dots \dots (i)$
(i) নং পরাবৃত্তের উপকেন্দ্রিক লম্বের সমীকরণ: $x = 4$ এবং উপকেন্দ্রিক লম্বের পাদবিন্দুর স্থানাঙ্ক = $(4, \pm 8)$



$$\begin{aligned} \therefore \text{ক্ষেত্রফল} &= 2 \int_0^8 (x_2 - x_1) dy = 2 \int_0^8 \left(4 - \frac{y^2}{16}\right) dy \\ &= 2 \left[4y - \frac{y^3}{48}\right]_0^8 = 2 \left[32 - \frac{8^3}{48}\right] \\ &= 2 \left[32 - \frac{32}{3}\right] = \frac{128}{3} \text{ বর্গ একক (Ans.)} \end{aligned}$$

02.

(গ) Solⁿ:

$$\begin{aligned} &\int_{-1}^1 (4 - p) \sqrt{p} dx \\ &= \int_{-1}^1 (4 - 4 + x^2) \sqrt{4 - x^2} dx \\ &= \int_{-1}^1 x^2 \sqrt{4 - x^2} dx \\ &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} (2 \sin \theta)^2 \sqrt{4 - (2 \sin \theta)^2} \cdot 2 \cos \theta d\theta \end{aligned}$$

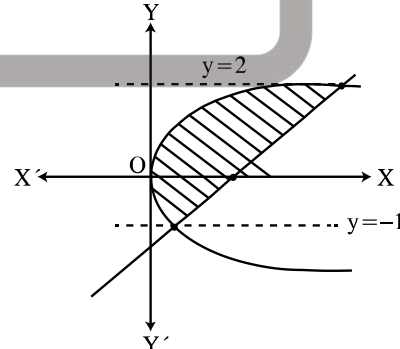
ধরি, $x = 2 \sin \theta$
 $\Rightarrow dx = 2 \cos \theta d\theta$

x	-1	1
θ	$-\frac{\pi}{6}$	$\frac{\pi}{6}$

$$\begin{aligned} &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 4 \sin^2 \theta \sqrt{4 - 4 \sin^2 \theta} 2 \cos \theta d\theta \\ &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 4 \sin^2 \theta \cdot 2 \cdot 2 \cos^2 \theta d\theta \\ &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 4 (2 \sin \theta \cos \theta)^2 d\theta = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 4 \sin^2 2\theta d\theta \\ &= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 2 (1 - \cos 4\theta) d\theta = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 2 d\theta - 2 \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos 4\theta d\theta \\ &= 2 \left[\theta\right]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} - 2 \times \frac{1}{4} [\sin 4\theta]_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \\ &= 2 \left(\frac{\pi}{6} + \frac{\pi}{6}\right) - \frac{1}{2} \times \left(\sin \frac{2\pi}{3} + \sin \frac{2\pi}{3}\right) \\ &= \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \text{ (Ans.)} \end{aligned}$$

03.

(গ) Solⁿ: দেওয়া আছে, $y^2 = x \dots \dots (i)$
 $y - x + 2 = 0 \dots \dots (ii)$





এখন, $y - y^2 + 2 = 0 \Rightarrow y^2 - y - 2 = 0$
 $\Rightarrow y^2 - 2y + y - 2 = 0 \Rightarrow (y + 1)(y - 2) = 0$
 $\therefore y = -1, 2$

$\therefore$ নির্ণেয় ক্ষেত্রফল

$$= \int_{-1}^2 [(y+2) - y^2] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2$$

$$= \frac{4}{2} + 4 - \frac{8}{3} - \frac{1}{2} - 2(-1) + \frac{(-1)^3}{3} \text{ বর্গ একক}$$

$$= 6 - \frac{8}{3} - \frac{1}{2} + 2 - \frac{1}{3} \text{ বর্গ একক} = \frac{9}{2} \text{ বর্গ একক (Ans.)}$$

04.

(খ) Solⁿ: দেওয়া আছে, $h(x) = \sin x$

প্রদত্ত রাশি $= \int_0^{\frac{\pi}{4}} \frac{\left\{ h\left(\frac{\pi}{2}-2x\right) \right\}^2}{\left\{ h\left(\frac{\pi}{2}-2x\right) \right\}^2 - \{h(2x)\}^2} dx$

$$= \int_0^{\frac{\pi}{4}} \frac{\left\{ \sin\left(\frac{\pi}{2}-2x\right) \right\}^2}{\left\{ \sin\left(\frac{\pi}{2}-2x\right) \right\}^2 - \{\sin 2x\}^2} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\cos^2 2x}{\cos^2 2x - \sin^2 2x} dx = \int_0^{\frac{\pi}{4}} \frac{\frac{1}{2} \cos 2x}{\cos 4x} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{1 + \cos 4x}{\cos 4x} dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{1}{\cos 4x} dx + \frac{1}{2} \int_0^{\frac{\pi}{4}} dx$$

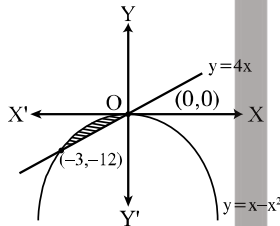
$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \sec 4x dx + \frac{1}{2} [x]_0^{\frac{\pi}{4}} = \frac{1}{8} [\ln |\sec 4x + \tan 4x|]_0^{\frac{\pi}{4}} + \frac{1}{2} \cdot \frac{\pi}{4}$$

$$= \frac{1}{8} [\ln |\sec \pi + \tan \pi| - \ln |\sec 0 + \tan 0|] + \frac{\pi}{8} = \frac{\pi}{8}$$

(Ans.)

05.

(গ) Solⁿ: দেওয়া আছে, $y = 4x$



আবার, $y = x - x^2 \Rightarrow 4x = x - x^2 \Rightarrow x^2 + 3x = 0$
 $\Rightarrow x(x + 3) = 0 \therefore x = 0, -3; y = 0, -12$
 $A_1 = \int_0^{-3} 4x dx = 18; A_2 = \int_0^{-3} x - x^2 dx = \frac{27}{2}$
ক্ষেত্রফল $= A_1 - A_2 = 18 - \frac{27}{2} = \frac{9}{2}$ বর্গ একক। (Ans.)

06.

(খ) Solⁿ: দেওয়া আছে, $f(x) = \sin x$

$\therefore f'(x) = \cos x \therefore \int_0^{\frac{\pi}{2}} \frac{f'(x)}{\{f(x)\}^2 - 16} \{f(x) - 3\} dx$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos x}{(\sin^2 x - 16)(\sin x - 3)} dx$$

ধরি, $z = \sin x$
 $dz = \cos x dx$

x	0	$\frac{\pi}{2}$
z	0	1

$$= \int_0^1 \frac{dz}{(z^2 - 16)(z - 3)}$$

$$= \int_0^1 \frac{dz}{(z-4)(z+4)(z-3)}$$

এখন $\frac{1}{(z-4)(z+4)(z-3)} \equiv \frac{A}{z-4} + \frac{B}{z+4} + \frac{C}{z-3}$
 $\Rightarrow 1 \equiv A(z+4)(z-3) + B(z-4)(z-3) + C(z-4)(z+4)$
 $+ C(z+4)(z-4)$

$z = 4$ বসিয়ে পাই, $1 = A \times 8 \times 1 \therefore A = \frac{1}{8}$

$z = 3$ বসিয়ে পাই, $1 = C \times 7(-1) \therefore C = \frac{-1}{7}$

$z = -4$ বসিয়ে পাই, $1 = B \times (-8)(-7) \therefore B = \frac{1}{56}$

$$\therefore \int_0^1 \frac{dz}{(z-4)(z+4)(z-3)} = \int_0^1 \left[\frac{1}{8(z-4)} + \frac{1}{56(z+4)} - \frac{1}{7(z-3)} \right] dz$$

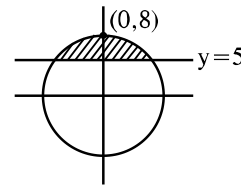
$$= \frac{1}{8} [\ln|z-4|]_0^1 - \frac{1}{56} [\ln|z+4|]_0^1 - \frac{1}{7} [\ln|z-3|]_0^1$$

$$= \frac{1}{8} \ln \left| \frac{-3}{-4} \right| + \frac{1}{56} \ln \left| \frac{5}{4} \right| - \frac{1}{7} \ln \left| \frac{-2}{-3} \right| = \frac{1}{8} \ln \left| \frac{4}{3} \right| + \frac{1}{56} \ln \left| \frac{4}{5} \right| - \frac{1}{7} \ln \left| \frac{2}{3} \right| \text{ (Ans.)}$$

07.

(গ) Solⁿ: দেওয়া আছে, $x^2 + y^2 = 64 \Rightarrow x^2 = 64 - y^2$

$\Rightarrow x = \pm \sqrt{64 - y^2}$



$\therefore$ নির্ণেয় ক্ষেত্রফল $= 2 \times \int_5^8 \sqrt{64 - y^2} dy$

ধরি, $y = 8 \sin \theta$

$\Rightarrow dy = 8 \cos \theta d\theta$

y	5	8
θ	$\sin^{-1} \frac{5}{8}$	$\frac{\pi}{2}$

$\therefore$ ক্ষেত্রফল $= 2 \times \int_{\sin^{-1} \frac{5}{8}}^{\frac{\pi}{2}} \sqrt{64 - 64 \sin^2 \theta} \cdot 8 \cos \theta d\theta$

$= 2 \times \int_{\sin^{-1} \frac{5}{8}}^{\frac{\pi}{2}} 8 \times \cos \theta \times 8 \cos \theta d\theta$

$= 64 \int_{\sin^{-1} \frac{5}{8}}^{\frac{\pi}{2}} 2 \cos^2 \theta d\theta = 64 \int_{\sin^{-1} \frac{5}{8}}^{\frac{\pi}{2}} (\cos 2\theta + 1) d\theta$

$= 64 \left[\frac{1}{2} \sin 2\theta + \theta \right]_{\sin^{-1} \frac{5}{8}}^{\frac{\pi}{2}}$

$= 64 \left[\frac{1}{2} \sin(\pi) + \frac{\pi}{2} - \frac{1}{2} \sin \left(2 \sin^{-1} \frac{5}{8} \right) - \sin^{-1} \left(\frac{5}{8} \right) \right]$

$= 64 \left[0 + \frac{\pi}{2} - \frac{1}{2} \times 2 \times \frac{5}{8} \times \frac{\sqrt{8^2 - 5^2}}{8} - \sin^{-1} \left(\frac{5}{8} \right) \right]$

$= 64 \left[\frac{\pi}{2} - \frac{5\sqrt{39}}{64} - \sin^{-1} \left(\frac{5}{8} \right) \right]$

$= 32\pi - 5\sqrt{39} - 64 \sin^{-1} \left(\frac{5}{8} \right) \text{ (Ans.)}$

08.

(ক) Solⁿ: ধরি, $I = \int \ln 2x dx = \int \ln 2x \cdot 1 dx$

$\ln 2x \int dx - \int \left[\frac{d}{dx} (\ln 2x) \int dx \right] dx$

$= (\ln 2x) \cdot x - \int \frac{1}{2x} \cdot 2 \cdot x dx$

$I = x \ln 2x - x + c$ এখন, $\int_1^2 \ln 2x dx = [x \ln 2x - x]_1^2$

$= (2 \ln 4 - 2) - (\ln 2 - 1) = 4 \ln 2 - 2 - \ln 2 + 1$

$\therefore \int_1^2 \ln 2x dx = 3 \ln 2 - 1 \text{ (Ans.)}$





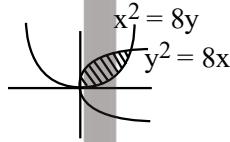
09.

(ক) Solⁿ: $\int_0^{\frac{\pi}{4}} \frac{dx}{1+\cos 2x} = \int_0^{\frac{\pi}{4}} \frac{dx}{2\cos^2 x} = \int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{2} dx$
 $= \frac{1}{2} [\tan x]_0^{\frac{\pi}{4}} = \frac{1}{2} [1 - 0] = \frac{1}{2}$ (Ans.)

(খ) Solⁿ: $\int x^2 p(x) dx = \int x^2 \cos x dx$
 $= x^2 \int \cos x dx - \int \left\{ \frac{d}{dx} (x^2) \int \cos x dx \right\} dx$
 $= x^2 \sin x - \int 2x \sin x dx$
 $= x^2 \sin x - 2 \left[x \int \sin x dx - \int \left\{ \frac{dx}{dx} \int \sin x dx \right\} dx \right]$
 $= x^2 \sin x - 2[-x \cos x - \int (-\cos x) dx]$
 $= x^2 \sin x + 2x \cos x - 2 \sin x + c$
 $\therefore \int_0^{\frac{\pi}{2}} x^2 p(x) dx$
 $= [x^2 \sin x + 2x \cos x - 2 \sin x]_0^{\frac{\pi}{2}}$
 $= \left(\frac{\pi^2}{4} - 2 - 0 \right) = \frac{\pi^2}{4} - 2$ (Ans.)

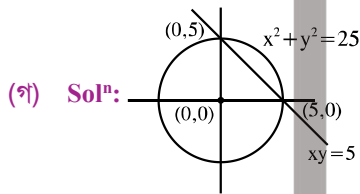
10.

(গ) Solⁿ: $g(x) = 0 \Rightarrow y^2 - 8x = 0 \Rightarrow y^2 = 8x \dots\dots\dots (i)$
 $h(x) = 0 \Rightarrow x^2 = 8y \dots\dots\dots (ii)$



(i) ও (ii) এর ছেদবিন্দুতে $y^4 = 64x^2 = 64 \times 8y$
 $\Rightarrow y(y^3 - 8^3) = 0 \Rightarrow y = 0$ অথবা $y = 8$
 $y = 0$ হলে $x = 0$ এবং $y = 8$ হলে $x = 8$
 $\therefore$ ক্ষেত্রফল $= \int_0^8 (y_2 - y_1) dx = \int_0^8 \left(\sqrt{8x} - \frac{x^2}{8} \right) dx$
 $= \left[\sqrt{8} \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{24} \right]_0^8 = \frac{128}{3} - \frac{64}{3} = \frac{64}{3}$ বর্গ একক। (Ans.)

11.



(গ) Solⁿ: দেওয়া আছে, $2x^2 + 2y^2 = 50 \Rightarrow x^2 + y^2 = 25 \dots\dots\dots (i)$
এবং $x + y = 5 \dots\dots\dots (ii)$
(ii) হতে y এর মান (i)-এ বসিয়ে পাই, $x^2 + (5-x)^2 = 25$
 $\Rightarrow x^2 + 25 - 10x + x^2 = 25$
 $\Rightarrow x^2 - 5x = 0 \therefore x(x-5) = 0$
 $\therefore x = 0, 5 \therefore y = 5, 0$
 $\therefore$ ক্ষুদ্রতর অংশের ক্ষেত্রফল $= \int_0^5 \{ \sqrt{25-x^2} - (5-x) \} dx$

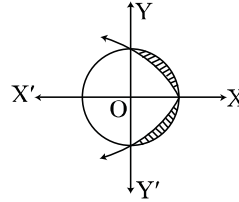
$$= \left[\frac{x}{2} \sqrt{25-x^2} + \frac{25}{2} \sin^{-1} \frac{x}{5} - 5x + \frac{x^2}{2} \right]_0^5$$

$$= \frac{5}{2} \sqrt{25-25} + \frac{25}{2} \sin^{-1} \frac{5}{5} - 25 + \frac{25}{2}$$

$$= \frac{25\pi}{4} - \frac{75}{2}$$
 বর্গ একক (Ans.)

12.

(গ) Solⁿ: $y^2 = 1 - x \Rightarrow y^2 = -(x-1)$ হতে মান
 $x^2 + y^2 = 1$



সমীকরণে বসিয়ে, $x^2 - (x-1) = 1 \Rightarrow x^2 + (1-x) = 1$
 $\Rightarrow x^2 - x = 1 - 1 \Rightarrow x(x-1) = 0$
 $\therefore x = 0$ অথবা, $x-1 = 0 \therefore x = 1 \therefore x = 0, 1$
 $x = 0$ হলে, $y^2 = 1 \therefore y = \pm 1$
 $x = 1$ হলে, $y^2 = 1 - 1 = 0 \therefore y = 0$
 $\therefore$ বক্ররেখাদ্বয়ের ছেদবিন্দু $(1,0), (0,-1), (0,1)$
এখন, x এর সীমা 0 হতে 1 পর্যন্ত $y_{up} = \sqrt{1-x^2}$ ও
 $y_{down} = -\sqrt{1-x^2}$ [x-অক্ষের উপরিভাগে]
বক্ররেখাদ্বয়ের সীমাবদ্ধ ক্ষেত্রফল $= 2 \int_0^1 (y_{up} - y_{down}) dx$
 $= 2 \int_0^1 (\sqrt{1-x^2} - (-\sqrt{1-x^2})) dx$
 $= 2 \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1} x + \frac{2}{3} (1-x)^{3/2} \right]_0^1$
 $= 2 \left(\frac{1}{2} \sin^{-1}(1) - \frac{2}{3} \right) = \sin^{-1}(1) - \frac{4}{3}$
 $= \left(\frac{\pi}{2} - \frac{4}{3} \right)$ বর্গ একক (প্রমাণিত)

13.

(ক) Solⁿ: দেওয়া আছে, $l(x) = \sin 2x \therefore \int l\left(\frac{x^\circ}{2}\right) dx$
 $= \int \sin \left(2 \times \frac{x^\circ}{2} \right) dx = \int \sin \frac{\pi x}{180} dx = -\frac{180}{\pi} \cos \frac{\pi x}{180} + c$

(খ) Solⁿ: দেওয়া আছে, $l(x) = \sin 2x$

$$\therefore \int \frac{\cos 2x}{(\sqrt{3+1(x)})^3} dx = \int \frac{\cos 2x dx}{(\sqrt{3+\sin 2x})^3}$$

$$\frac{1}{2} dz = \cos 2x dx ; \int \frac{\frac{1}{2} dz}{(\sqrt{z})^3} = \frac{1}{2} \int z^{-\frac{3}{2}} dz = \frac{1}{2} \frac{z^{-\frac{1}{2}}}{-\frac{1}{2}}$$

$$= -z^{-\frac{1}{2}} = -\frac{1}{\sqrt{3+\sin 2x}} + c$$
 (Ans.)

(গ) Solⁿ: দেওয়া আছে, $g(x) = \log x$

$$\therefore \int \frac{1}{x(1+g(x))^3} dx = \int \frac{dx}{x(1+\log x)^3}$$

ধরি, $1 + \log x = z \Rightarrow \frac{dx}{x} = dz$

এখন, $\int \frac{dz}{z^3} = \frac{z^{-3+1}}{-3+1} + c = -\frac{1}{2} \times \frac{1}{z^2} + c = \frac{-1}{2(1+\log x)^2} + c$

